Quantum Gravity Dynamics

Chapter 4: Natural-Invariant Dynamics
and the Physical Meaning of Gravity

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Abstract

What does the natural-invariant description of gravitation add to the Einsteinian metric equations? We formulate a precise first answer, without postulating a new wave law. Given the composite QGD metric of Chapters 1–2 and the effective gravitational and matter action used in Chapters 2–3, variation with respect to the independent natural-invariant one-forms produces an exact projected Euler equation. The historical three-channel idea thereby admits an unambiguous *action-defined successor*, $\mathscr{S} = \mathscr{G} + \mathscr{T} + \mathscr{Q} = 0$. In the derivative-free, metric-only sector, $\mathscr{Q} = 0$ and the remaining terms are projections of the Einstein–matter residual. On a regular full-rank signed-field representation, those projected equations are equivalent to the effective Einstein equations. The result clarifies the physical role of σ_μ, the origin of its representation freedom, and the exact location where genuinely distinct QGD dynamics would have to arise. We incorporate the Chapter 3 breakthroughs: Newtonian calibration of the exterior natural-invariant amplitude; the exact static Killing-energy expansion; a corrected conserved-current bridge; the leading Dirac-derived quantum density-gradient stress; and a no-go theorem for universal power-law exterior corrections produced solely by restructuring a compact spherical source at fixed mass. It does not assert a microscopic wave equation, a derived σ kinetic term, or a new gravitational observable. This short chapter is intended to grow as these questions are solved; it does not alter the historical Chapter 4. The chapter proves a metric-only quantum-equivalence theorem, imports a verified semiclassical massless-Dirac far-field correction, and audits the historical 9/2 force. A QGD–110 section derives the quantum structure of the master metric from the founding principles: a universal quantum sector fixed by the source's quantum stress, a per-wave quantum conformal factor, and the test of the historical derivative term, and the exact flow law of spin-$\frac{1}{2}$ matter in a curved universal metric; it also records Lorentz-covariance and gauge tests of the main equations. A QGD–115 subsection shows that the Einstein–Hilbert form of the gravitational dynamics is not an independent axiom: it follows uniquely, by Lovelock's theorem, from QGD's metric-only structure and a second-order locality principle, with the coupling fixed by the Newtonian calibration. A QGD–120 subsection states the complete QGD action, its variables, its symmetries and its quantum effective form; a QGD–121 subsection shows that on spanning branches it propagates exactly the two gravitational modes of general relativity. The revision history is in Section 16. In the sense stated in Chapter 1's introduction, QGD is a theory of quantum gravity in the classical/emergent branch: quantum matter sources a classical geometry, the quantum structure of that geometry is derived here, and the metric itself is not quantized.

1 Q061 final audit of the covariant QGD action and its extensions (5 November 2026)

1.1 Provenance: two axioms, one separate dynamical law

The founding geometric postulate is the full $g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}]$. Axiom II postulates compatible microscopic realizability but fixes no classical or quantum action coefficient. The second-order, local, generally covariant *metric-mediated* gravitational law used here is selected *conditionally* on Principle L; four-dimensional Lovelock uniqueness then gives the Einstein tensor and a cosmological constant, while Newtonian matching fixes G_N. This theorem does not exclude an additional direct-sector functional. The metric-only operational model always declares $\lambda\mathcal{Q} = 0$ and restricts to a suitable regular field-map branch.

1.2 Renormalization-conscious effective action

For clarity, in units $\hbar = c = 1$ except as a loop-order annotation, define a once-subtracted renormalized metric functional by

$$\begin{aligned}\Gamma^{\rm ren}_{\rm met}[g, \psi_c, A_c; \varrho, \mu] = {} & \frac{1}{16\pi G_N} \int_{\mathcal{M}} d^4x \sqrt{-g}(R - 2\Lambda_{\rm ren}) + S_{\rm GHY}[g] + S^{\rm cl}_{\rm F1}[\psi_c, A_c; g] \\ & + \int_{\mathcal{M}} d^4x \sqrt{-g} \left[c_R(\mu) R^2 + c_C(\mu) C_{\alpha\beta\gamma\delta} C^{\alpha\beta\gamma\delta} + c_E(\mu) E_4 \right] \\ & + \Gamma^{\rm ren}_{\rm matter,rem}[g, \psi_c, A_c; \varrho, \mu] + \Gamma_{\rm higher/bdy}.\end{aligned} \tag{1}$$

Set $\Gamma^{\rm ren}_{\rm QGD}[\Sigma, \ldots] = \Gamma^{\rm ren}_{\rm met}[g(\Sigma), \ldots]$. The c_i are independent finite-renormalized effective coefficients (including the matched quantum local parts) rather than guaranteed overall-order-$\hbar$ parameters. QN004 computes particular Euclidean one-loop form factors; the remainder in Eq. (1) explicitly excludes local pieces already placed in c_i. E_4 is topological in four dimensions for fixed topology up to necessary boundary qualifications. Higher-derivative terms are treated perturbatively inside their EFT domain, not as a license to retain all formal runaway branches.

1.3 Symmetries and the external-coframe qualification

Under simultaneous transformations of all geometric fields, the metric-only action is spacetime-diffeomorphism covariant and local-frame Lorentz invariant; the Dirac spin connection and common electromagnetic $F_{\mu\nu}$ preserve spinor-frame covariance and local $U(1)$. For Y the $N \times 4$ sector matrix, $Y \mapsto O(x)Y$ with $O^{\mathsf{T}} C O = C$ leaves g unchanged when $\lambda\mathcal{Q} = 0$. These metric-preserving directions give no independent metric-only energy, even when $O(x)$ is nonconstant. In the declared variational scheme, however, the reference coframe T is *held fixed*. Formal covariance with T transforming as a coframe does not itself establish unrestricted off-shell active diffeomorphism gauge freedom; the Noether identity must include any external reference background. An independently derivative-dependent $\mathcal{Q}$ requires its own covariant tensorial prescription and variational adjoint before it can be used.

1.4 Causal quantum response and the additional-sector option

For time-dependent expectation values use a Schwinger–Keldysh closed-time-path functional $\Gamma_{\rm CTP}[\Sigma_+, \Sigma_-; \varrho_{\rm in}]$, varying its forward history and then taking the physical limit $\Sigma_+ = \Sigma_-$. Euclidean QN004 response functions cannot simply be substituted as retarded real-time kernels without an analytic/state prescription. A direct dark-energy functional has the separate form

$$\Gamma_{\rm ext} = \Gamma^{\rm ren}_{\rm QGD} + \Delta S_{\rm sector}[\Sigma; \mathsf{H}_0, \text{parameters}], \qquad \frac{\delta\Gamma_{\rm ext}}{\delta\Sigma} = (D_\Sigma g)^* \frac{\delta\Gamma_{\rm ext}}{\delta g} + (D_\Sigma s)^* \frac{\delta\Gamma_{\rm ext}}{\delta s} \tag{2}$$

when the new functional depends through $g(\Sigma)$ and an independent selected scalar $s(\Sigma)$. If the joint map (g, s) has a right inverse on the admissible variation spaces, the full sector Euler equation is equivalent to the two separate metric and scalar Euler equations. The additional selector/reference, kinetic coefficient, and potential scale are new physical inputs. Signed sectors and universal metric coupling alone cannot determine them (Q055–Q060). Thus Eq. (1), not Eq. (2), remains the canonical action.

The defining QGD action and the two-axiom hierarchy (5 November 2026)

Action placement. Chapter 1 states Axiom I (complete master metric), Axiom II (unproved independent microscopic realizability), and the separate defining variational law at the outset. This chapter is the central *dynamical derivation*, not a deferred search for a law of gravity. The classical dynamics is the local Einstein–Hilbert–matter effective action expressed in independent natural-invariant variables, with renormalized matter loops added consistently:

$$\boxed{\begin{aligned}\Gamma^{\mathrm{ren}}_{\mathrm{QGD}}[\Sigma,\psi_c,A_c;\varrho,\mu] = \frac{1}{16\pi G_N}\int_{\mathcal{M}} d^4x\sqrt{-g(\Sigma)}\,[R(g(\Sigma))-2\Lambda_{\mathrm{ren}}] \\ + S_{\mathrm{GHY}}[g(\Sigma)] + S^{\mathrm{cl}}_{\mathrm{F1}}[\psi_c,A_c;g(\Sigma)] \\ + \hbar\Gamma^{(1)}_{\mathrm{m,ren}}[g(\Sigma),\psi_c,A_c;\varrho,\mu] + O(\hbar^2).\end{aligned}} \tag{3}$$

Here $S_{\mathrm{GHY}} = (8\pi G_N)^{-1}\int_{\partial\mathcal{M}} \epsilon\sqrt{|h|}\,K$ with the standard boundary prescription, and

$$\boxed{S^{\mathrm{cl}}_{\mathrm{F1}} = \int d^4x\sqrt{-g}\left[\frac{i}{2}\bar\psi_c\gamma^\mu \overleftrightarrow{\nabla}_\mu\psi_c - m_0\bar\psi_c\psi_c + G_4(\bar\psi_c\psi_c)^2 - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right].} \tag{4}$$

The quantum term is decomposed in one renormalization scheme into local $\alpha_R R^2 + \alpha_C C^2 + \alpha_E E_4$, complementary nonlocal kernels, state contributions and mixed vertices. Chapter 6, Section 40, gives explicit one-loop Euclidean massive Dirac/Hartree form factors through curvature-quadratic order, not an arbitrary-curvature Lorentzian completion. These contributions must not be appended twice.

Full-master equations and declared specialization.

$$\boxed{2s_s\Pi^{ab}\sigma^{(s)}_b = \lambda[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi]^a, \qquad \Pi^{ab} = T^a{}_\mu T^b{}_\nu \frac{1}{\sqrt{-g}}\frac{\delta\Gamma_{\mathrm{QGD}}}{\delta g_{\mu\nu}}.} \tag{5}$$

This is the action's chain-rule equation for fixed reference coframe and without additional direct Σ dependence, provided $\mathcal{Q}$ is defined as a differentiable functional with its boundary prescription. In the *defined* metric-only specialization $\lambda\mathcal{Q} = 0$, it reduces to $\Pi^{ab}\sigma^{(s)}_b = 0$. A full-rank metric Jacobian implies $\Pi^{ab} = 0$, while rank-deficient sectors can be spuriously stationary. Axiom II is not used to establish this result. If an independent direct sector action were added, its Euler derivative would modify this equation; no such addition is adopted here.

Interpretive restriction. Lovelock uniqueness motivates the local classical metric dynamics only after the hypotheses of locality, four dimensions and second-order metric Euler equations are imposed. It does not prove that this particular variational action follows from the geometric axioms alone. The present metric-only theory is physically equivalent to the corresponding metric EFT on regular complete branches unless additional *specified and justified* sector dynamics or admissibility conditions yield distinct observables.

Reading key. **D**: derived from stated assumptions; **A**: adopted effective dynamics; **C**: fixed definition or normalization; **P**: proposed physical interpretation not yet an independent empirical result. These labels distinguish logical status, not scientific importance.

2 From the quantum description to effective geometry

General relativity establishes how geometry behaves when coupled to energy and momentum. It does not, by itself, establish whether geometry is fundamental or an effective macroscopic description. QGD investigates a concrete alternative interpretation: the gravitational field is associated with macroscopic quantum structure of matter, represented by gravitational natural-invariant covectors $\sigma_\mu^{(s)}$; their composites define an effective metric (P). The word *natural invariant* is QGD's historical name for these covectors; a covector is not itself a coordinate-invariant scalar.

The essential distinction is between an *ontological* proposal and a *dynamical* result. The first three chapters supply (i) the four-component macroscopic spinor structure and controlled local spinor ratios, (ii) the master metric and its signed-sector representations, and (iii) Einsteinian effective dynamics and Newtonian calibration in specified limits. None of these statements alone proves that the individual natural-invariant components constitute independent observable particles or waves.

The complete founding QGD master metric is retained unchanged (the signed-sector-only formula is a specialization, not the master metric):

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_{s=1}^{N} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \quad \varepsilon_s = \pm 1.} \tag{6}$$

Here $T^a{}_\mu$ is the reference coframe, and the derivative operator $\mathcal{Q}_{ab}$ and coefficient λ remain unmatched proposals in the unrestricted theory. We will derive definite equations in the explicitly stated fixed-coframe, $\lambda = 0$, metric-only effective sector and then explain what changes when the full metric map is used.

3 The new Chapter 4 field equation

In this section set $c = 1$. Restore the physical Einstein coupling $8\pi G_N/c^4$ when using SI units. Consider the composite effective action (A)

$$\Gamma[\sigma, \Psi] = S_{\rm EH}[g(\sigma)] + S_{\rm matter}[g(\sigma), \Psi], \qquad S_{\rm EH} = \frac{1}{16\pi G_N} \int \mathrm{d}^4x \sqrt{-g}\,(R - 2\Lambda). \tag{7}$$

Boundary variations are compactly supported or supplied with the appropriate gravitational boundary term. The matter equations are varied independently. Define $T^{\mu\nu}$ by $\delta S_{\rm matter} = \frac{1}{2}\int \sqrt{-g}\, T^{\mu\nu}\delta g_{\mu\nu}$ for on-shell matter variations, and let

$$\boxed{\mathcal{E}_g^{\mu\nu} := \frac{1}{\sqrt{-g}} \frac{\delta \Gamma}{\delta g_{\mu\nu}} = -\frac{1}{16\pi G_N}\left(G^{\mu\nu} + \Lambda g^{\mu\nu} - 8\pi G_N T^{\mu\nu}\right).} \tag{8}$$

This is a definition of the effective metric Euler derivative, not an independent QGD field equation.

3.1 Why the Einstein–Hilbert form: a uniqueness theorem (QGD–115)

The action (7) was introduced as an adopted effective action (A). Its *form* need not be adopted. It follows from facts established within QGD together with one structural principle.

Established within QGD.

(i) *Matter couples only through g.* The microscopic action uses a vierbein of g and is locally Lorentz invariant, so matter depends on the natural invariants only through g (Proposition 13.6).

(ii) *Covariance.* The master metric is tensorial under coordinate changes and invariant under frame Lorentz transformations (Section 13).

(iii) *Conservation.* Diffeomorphism invariance of the matter action gives $\nabla_\mu T^{\mu\nu} = 0$ on shell.

(iv) *No consistent dynamics beyond g has been found.* The routes tested, physical frame redundancies, an antisymmetric pairing field and two dynamical frames, were either inconsistent at two-derivative order or required an independent postulate (Section 14).

Principle L (locality and order). The gravitational equation for the natural invariants is local, generally covariant, depends on σ only through g, and contains at most second derivatives of g.

Theorem 3.1 (Einstein form by uniqueness (D|L; Lovelock))**.** *Under (i)–(iii) and Principle L, in four spacetime dimensions the gravitational equation has the form $\mathcal{E}_{\mu\nu}[g] = \kappa T_{\mu\nu}$ with $\mathcal{E}_{\mu\nu}$ symmetric, built from g and its first two derivatives, and identically divergence free. By Lovelock's theorem [12, 13] the only such tensors are $aG_{\mu\nu} + bg_{\mu\nu}$, so*

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.} \tag{9}$$

The Newtonian calibration of Chapter 3 then fixes $\kappa = 8\pi G_N/c^4$, and the action (7) is the corresponding variational form, up to boundary terms.

Proof. The equation must be divergence free because its source is, by (iii), for every matter configuration. Lovelock's theorem classifies the symmetric, divergence-free concomitants of g with at most second derivatives in four dimensions. The coupling is the single constant left, and F5 fixes it. □

Verification. On a generic metric with all ten components varying, a 50-digit computation gives $\nabla^\mu G_{\mu\nu} = O(10^{-14})$, while $\nabla^\mu R_{\mu\nu}$ and $\nabla^\mu(Rg_{\mu\nu})$ are of order 0.1–0.7; the unique divergence-free combination $aR_{\mu\nu} + bRg_{\mu\nu}$ has $b/a = -0.5000$.

Status. The *form* of the gravitational dynamics is derived (D|L), and the *coupling* is fixed by the calibration F5. Three things are not derived:

- **Λ.** The theorem allows a cosmological term but does not fix its value.
- **The second-order requirement.** It is a principle, chosen for a well-posed, ghost-free classical limit. Higher-derivative terms, such as the logarithmic curvature-squared terms induced by quantum matter, are treated as subleading corrections.
- **The value of G_N from the microscopic theory.** Two routes remain open. In the induced-gravity route (QGD–114), the one-loop effective action of Chapter 1's fields yields the Einstein form, but with one spin-$\frac{1}{2}$ field and one Maxwell field the induced Newton constant is negative and the induced cosmological constant is of cutoff order [15, 16]. In the thermodynamic route [14], horizon entropy counts physical degrees of freedom positively and Λ appears as an integration constant, but an equilibrium hypothesis and a physical cutoff are required.

Write the metric-only signed specialization locally as

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \sum_s s_s \sigma^{(s)}_\mu \sigma^{(s)}_\nu, \qquad s_s := -\varepsilon_s \in \{+1, -1\}. \tag{10}$$

All variations below treat the covectors as independent components in a fixed frame and hold the background coframe fixed.

Theorem 3.2 (Action-defined natural-invariant equation (D|A))**.** *For the action* (7) *and metric map* (10)*, the complete Euler derivative of each independent natural-invariant covector satisfies*

$$\boxed{\mathscr{S}^{\rho}_{(s)} := \frac{1}{\sqrt{-g}}\frac{\delta\Gamma}{\delta\sigma^{(s)}_{\rho}} = 2s_s\mathcal{E}_g^{\rho\nu}\sigma^{(s)}_{\nu} = 0.} \tag{11}$$

Equivalently, with action-defined *channels,*

$$\boxed{\mathscr{S}^{\rho}_{(s)} = \underbrace{-\frac{s_s}{8\pi G_N}\left(G^{\rho\nu}+\Lambda g^{\rho\nu}\right)\sigma^{(s)}_{\nu}}_{\mathscr{G}^{\rho}_{(s)}} + \underbrace{s_s T^{\rho\nu}\sigma^{(s)}_{\nu}}_{\mathscr{T}^{\rho}_{(s)}} + \underbrace{0}_{\mathscr{Q}^{\rho}_{(s)}} = 0.} \tag{12}$$

Proof. Vary the signed metric while holding other sectors fixed: $\delta g_{\mu\nu} = s_s(\delta\sigma^{(s)}_{\mu}\sigma^{(s)}_{\nu} + \sigma^{(s)}_{\mu}\delta\sigma^{(s)}_{\nu})$. Symmetry of $\mathcal{E}_g^{\mu\nu}$ gives $\delta\Gamma = \int\sqrt{-g}\,2s_s\mathcal{E}_g^{\rho\nu}\sigma^{(s)}_{\nu}\delta\sigma^{(s)}_{\rho}$. Equation (11) follows from independent arbitrary compactly supported variations. Substituting (8) proves (12). □

Remark 3.3 (What the three symbols mean)**.** *$\mathscr{G}$ is the effective* geometric-action contribution*, $\mathscr{T}$ is the* matter-stress contribution*, and $\mathscr{Q}$ denotes additional terms arising from a* specified *derivative-dependent metric map when such a map has actually been defined and varied. Thus $\mathscr{Q} = 0$ in the metric-only algebraic sector. The letter Q does* not *by itself mean "quantum". In particular, a genuine quantum matter-loop correction to a metric-only effective action modifies $\mathcal{E}_g$, hence the geometric/matter balance, without necessarily producing any separate $\mathscr{Q}$ channel.*

3.2 The complete QGD action (QGD–120)

The action (7) left S_{matter} generic. Every ingredient needed to make it explicit has been derived in Chapters 1–4; here it is assembled.

Fields. The dynamical fields are the natural invariants $\sigma^{(s)}_a$ $(s = 1, \ldots, N$, signs $s_s = \pm 1)$, the microscopic spin-$\frac{1}{2}$ field ψ and the Maxwell field A_μ. The reference coframe $T^a{}_\mu$ is held fixed. The metric is the classical restriction of the master metric,

$$g_{\mu\nu} = T^a{}_\mu T^b{}_\nu\Big[\eta_{ab} + \sum_s s_s\sigma^{(s)}_a\sigma^{(s)}_b\Big]. \tag{13}$$

There is no independent derivative term $\lambda\mathcal{Q}$. The quantum content of the master metric arises from the effective action below, not from an added term (Section 13).

Definition 3.4 (QGD action)**.**

$$\boxed{S_{\text{QGD}}[\sigma,\psi,A] = \frac{1}{16\pi G_N}\int_{\mathcal{M}}\mathrm{d}^4x\sqrt{-g}\,(R-2\Lambda) + \frac{1}{8\pi G_N}\oint_{\partial\mathcal{M}}\mathrm{d}^3y\,\epsilon\sqrt{|h|}\,K + S_{\text{F1}}[\psi,A;g]} \tag{14}$$

with $g = g[\sigma;T]$ from (13)*, the Gibbons–Hawking–York term for a well-posed variational principle, and the microscopic action of Chapter 1 written on g,*

$$S_{\text{F1}} = \int\mathrm{d}^4x\sqrt{-g}\Big[\tfrac{i}{2}\big(\bar\psi\gamma^\mu\nabla_\mu\psi - \overline{\nabla_\mu\psi}\gamma^\mu\psi\big) - m_0\bar\psi\psi + G_4(\bar\psi\psi)^2 - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}\Big]. \tag{15}$$

Here G_4 is Chapter 1's contact coupling (renamed to avoid a clash with G_N), $\gamma^\mu = e_a{}^\mu\gamma^a$ with e any vierbein of g, $\{\gamma^\mu,\gamma^\nu\} = -2g^{\mu\nu}$, and $\nabla_\mu = \partial_\mu + \Gamma_\mu + iqA_\mu$ with the spinor connection fixed by $\nabla\gamma = 0$.

Theorem 3.5 (Symmetries of the QGD action (D))**.** *S_{QGD} is invariant under:*

1. *diffeomorphisms acting on all fields, T included;*
2. *local Lorentz transformations of the frame index, $T \mapsto \Lambda T$, $\sigma_a \mapsto \sigma_c(\Lambda^{-1})^c{}_a$, together with local Lorentz rotations of the vierbein e and the spinor ψ;*
3. *local sector rotations $\sigma \mapsto O\sigma$ with $O^{\mathsf{T}}CO = C$, $C = \mathrm{diag}(s_s)$;*
4. *local $U(1)$ gauge transformations of (ψ, A).*

Any configuration with $\sum_s s_s \sigma_a^{(s)} \sigma_b^{(s)} = 0$ and flat T gives Minkowski space, whose isometry group is the Poincaré group. For perturbation theory the vacuum must be represented on a spanning branch, not at $\sigma = 0$ (Theorem 4.4).

Proof. The gravitational terms depend on (σ, T) only through g, which is invariant under 2 and 3 and a tensor under 1 (Section 13, covariance table). The matter term depends on g through a vierbein; its local Lorentz invariance follows from $\nabla\gamma = 0$, under which $[\nabla_\mu, \nabla_\nu] = -\frac{1}{4} R_{ab\mu\nu}\gamma^a\gamma^b$ (verified in Chapter 4, Theorem 13.7). The $U(1)$ invariance is standard. □

Variables and the role of T. S_{QGD} is varied with respect to σ, ψ and A; T is not varied. This does not break covariance in effect. On any branch where the metric map has a right inverse, every metric variation is produced by a variation of σ (Chapter 2, Correspondence Theorem), so a change of T can be compensated by σ, and the physical content is independent of the choice of T. The individual $\sigma^{(s)}$ are therefore not gauge-invariant observables: only g and quantities built from it are (Section 13).

Field equations. Varying σ gives the field-balance equation (11). On spanning branches it is equivalent to $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu}$, whose form is unique by Theorem 3.1. Varying ψ gives the macroscopic-limit spinor equation on g; varying A gives Maxwell's equations with the $U(1)$ current.

Verification. For the static branch $\sigma = A(r)\,\mathrm{d}t + B(r)\,\mathrm{d}r$ on the flat spherical reference, the Euler–Lagrange equations of $\sqrt{-g}\,R$ computed directly in sector variables are exactly twice those of the reduced Lagrangian $(rF)'/\sqrt{D} + \sqrt{D}$ used in Chapters 2–3, for both A and B. That reduced Lagrangian is therefore the Einstein–Hilbert action in natural-invariant variables, up to normalization and a total derivative.

Quantum effective action. At the quantum level the matter fields are integrated out on $g(\sigma)$:

$$\begin{aligned} \Gamma_{\mathrm{QGD}}[\sigma] &= \frac{1}{16\pi G_N}\int \sqrt{-g}\,(R - 2\Lambda) + S_{\mathrm{GHY}} + \Gamma_{\mathrm{F1}}[g(\sigma)], \\ e^{i\Gamma_{\mathrm{F1}}[g]/\hbar} &= \int \mathcal{D}\bar{\psi}\,\mathcal{D}\psi\,\mathcal{D}A\; e^{iS_{\mathrm{F1}}[\psi, A; g]/\hbar}. \end{aligned} \tag{16}$$

with gauge fixing and a regulator; F2's coarse-graining is the Wilsonian version of Γ_{F1}. Stationarity of Γ_{QGD} gives the semiclassical equations with $\langle T_{\mu\nu}\rangle$. The quantum structure derived in Section 13, namely the universal sector q, the per-wave factor Ω_ψ, and the spin-$\frac{1}{2}$ flow law, consists of consequences of (16), not additional terms.

Status.

Element	Status
Einstein–Hilbert form	Derived under Principle L (Theorem 3.1)
G_N	Fixed by the Newtonian calibration (Chapter 3)
Λ	Allowed, not determined
Matter sector S_{F1}	Chapter 1's microscopic action, covariantly coupled
Coupling of matter to σ	Only through g (Proposition 13.6)
Derivative term $\lambda\mathcal{Q}$	Absent; the historical trial is excluded (Proposition 13.3)
Symmetries	Diffeomorphisms, local Lorentz, local sector rotations, $U(1)$
Quantum level	Γ_{QGD}: matter quantized on a classical $g(\sigma)$
Quantization of σ itself	Not defined: it would require a measure for σ and gauge fixing of the sector redundancy, and is not needed under the semiclassical reading

In summary, S_{QGD} is a well-defined, generally covariant and locally Lorentz-invariant classical action, and Γ_{QGD} is its well-defined semiclassical quantum effective action. A path integral over the natural invariants themselves is not part of QGD as formulated.

3.3 Renormalized final action and quantum-to-classical limit (QGD–136)

Scope and status. This subsection consolidates the Chapter 4 action with the conditional static matter-wave correspondence of Chapter 5 v3.0 and the massless-Dirac semiclassical correction of Chapter 5 v3.1. It is a proposed *effective, semiclassical* final form, not a derivation of an autonomous collective metric from Chapter 1's flat-space microscopic theory. We work with $c = 1$ and display $\hbar$ as a formal loop-counting parameter. To discuss a classical limit, hold fixed the *renormalized* G_N, the macroscopic classical matter source and appropriate boundary data; a naive $\hbar \to 0$ limit of an arbitrarily normalized microscopic spinor is not the same limit.

Definition 3.6 (Quantum-completed QGD effective action (EFT))**.** *For the explicitly controlled and currently accepted semiclassical branch, set $\lambda\mathcal{Q}_{ab} = 0$ in the as-yet-unmatched historical metric-derivative proposal, and take the reduced classical metric map*

$$g_{\mu\nu}[\sigma; T] = T^{a}{}_{\mu} T^{b}{}_{\nu} \left[\eta_{ab} + \sum_{s} s_s \sigma_a^{(s)} \sigma_b^{(s)} \right]. \tag{17}$$

On a regular signed-sector branch, define the renormalized *semiclassical action, with a specified quantum matter state ϱ,*

$$\boxed{\begin{aligned} \Gamma^{\mathrm{ren}}_{\mathrm{QGD}}[\sigma, \psi_c, A_c; \varrho] = {} & \frac{1}{16\pi G_N} \int_{\mathcal{M}} \mathrm{d}^4x \sqrt{-g(\sigma)}\, [R(g(\sigma)) - 2\Lambda_{\mathrm{ren}}] \\ & + S_{\mathrm{GHY}}[g(\sigma)] + S^{\mathrm{cl}}_{\mathrm{F1}}[\psi_c, A_c; g(\sigma)] \\ & + \hbar\, \Gamma^{(1)}_{\mathrm{m,ren}}[g(\sigma), \psi_c, A_c; \varrho; \mu] + O(\hbar^2). \end{aligned}} \tag{18}$$

Here ψ_c, A_c are mean/classical matter backgrounds where retained. The quantum functional includes one-loop effects of fluctuating matter fields not already present in the classical background action. *The matter action $S^{\mathrm{cl}}_{\mathrm{F1}}$ is precisely Eq. (15), with its spin connection, Maxwell and four-fermion components. When* all *matter is integrated out rather than retaining ψ_c, A_c, one uses the equivalent background-free quantum influence functional and must not also count the same matter stress in $S^{\mathrm{cl}}_{\mathrm{F1}}$. The finite one-loop functional is defined using a consistent regulator, subtraction scheme, renormalization scale and appropriate boundary/state prescription. A finite-cutoff interpretation is allowed for Chapter 1's four-fermion effective interaction.*

What the symbol $\Gamma^{(1)}_{\mathrm{m,ren}}$ contains. The quantum term of Eq. (18) is *not* merely a chosen local correction to the natural-invariant metric. In four dimensions its renormalized curvature-dependent part has schematically

$$\begin{aligned}\Gamma^{(1)}_{\mathrm{m,ren}} = &\int \mathrm{d}^4x\sqrt{-g}\left[a_R(\mu)R^2 + a_C(\mu)C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}\right]\\ &+\Gamma^{(1)}_{\mathrm{nonlocal}}[g;\mu] + \Gamma^{(1)}_{\mathrm{state}}[g;\varrho] + \Gamma^{(1)}_{\mathrm{vertices}}[g,\psi_c,A_c;\varrho],\end{aligned} \tag{19}$$

with local curvature-squared renormalization and, for example, nonlocal curvature kernels of the form $\int\sqrt{-g}\,C\log(-\Box/\mu^2)C$ and $\int\sqrt{-g}\,R\log(-\Box/\mu^2)R$ when justified by the relevant massless-field calculation. The coefficients and admissible kernels are theory-, state- and scheme-dependent and are *not* freely fixed to reproduce a desired force. Topological and boundary counterterms may be needed and are understood in the chosen variational problem. Equations (18)–(19) are an organizing form, *not* a claim that Chapters 1–5 have calculated every one-loop form factor. For time-dependent semiclassical backreaction the appropriate object is a causal in–in (Schwinger–Keldysh) effective action; an in–out determinant is not generally a causal evolution law.

Proposition 3.7 (Controlled Einstein limit, D|A)**.** *Suppose (i) $\lambda\mathcal{Q}$ is absent or tends to zero, (ii) any independent classical R^2, C^2 and other higher-curvature couplings are set to zero at the selected classical matching point, (iii) the renormalized G_N and Λ remain finite in the limit, (iv) the matter state admits a smooth classical macroscopic stress limit, and (v) all loop/higher-order remainders in Eq. (18) vanish as $\hbar\to 0$ on the specified regular branch. Then the full-rank QGD natural-invariant variation reduces to the Einstein–matter equations,*

$$\boxed{\lim_{\hbar\to 0}\frac{\delta\Gamma^{\mathrm{ren}}_{\mathrm{QGD}}}{\delta\sigma^{(s)}_{\rho}} = 0 \quad\Longleftrightarrow\quad G_{\mu\nu}+\Lambda g_{\mu\nu} = 8\pi G_N T^{\mathrm{cl}}_{\mu\nu}.} \tag{20}$$

Proof. The quantum term and higher-order remainders vanish by hypotheses (iv)–(v). The remaining action is the Einstein–Hilbert–GHY plus classical matter action at fixed renormalized couplings. Its metric Euler tensor is $\mathcal{E}_g^{\mu\nu}\propto G^{\mu\nu}+\Lambda g^{\mu\nu}-8\pi G_N T^{\mu\nu}_{\mathrm{cl}}$. The composite metric variation is $\delta g_{\mu\nu} = 2\sum_s s_s\sigma^{(s)}_{(\mu}\delta\sigma^{(s)}_{\nu)}$. Hence stationarity in the sector variables gives $2s_s\mathcal{E}_g^{\rho\nu}\sigma^{(s)}_{\nu} = 0$. Surjectivity of the metric map on a regular signed-sector branch makes this equivalent to $\mathcal{E}_g^{\mu\nu}=0$. The claimed classical Einstein equations follow. □

Where Chapter 5's $g_c^2/(4\pi Z)$ enters (not an extra kinetic term). The established *conditional* static collective-field model of Chapter 5 v3.0 uses $\mathcal{E}[q_{\mathrm{coll}};\rho_m] = \int\mathrm{d}^3x\,[\frac{Z}{2}|\boldsymbol{\nabla}q_{\mathrm{coll}}|^2 - g_c q_{\mathrm{coll}}\rho_m]$, $Z>0$. Its result

$$\boxed{G_{\mathrm{eff}} = \frac{g_c^2}{4\pi Z}, \qquad G_{\mathrm{eff}}\overset{\text{calibration}}{=}G_N, \qquad H = \frac{2g_c q_{\mathrm{coll}}}{c^2} = -\frac{2\Phi_N}{c^2}} \tag{21}$$

provides a *static infrared matching condition* on the coefficient already appearing in the Einstein–Hilbert term. Under this change of variables the collective energy becomes

$$\boxed{\mathcal{E}_H = \int\mathrm{d}^3x\left[\frac{c^4}{32\pi G_{\mathrm{eff}}}|\boldsymbol{\nabla}H|^2 - \frac{c^2}{2}\rho_m H\right], \qquad \nabla^2 H = -\frac{8\pi G_{\mathrm{eff}}}{c^2}\rho_m.} \tag{22}$$

Here q_{coll} denotes Chapter 5's collective scalar coordinate, not Chapter 4's distinct quantum metric deformation q_{ab}. This equality is exact *inside the stated static scalar effective model*; it is not a derivation of its mode from the original microscopic Dirac action or of the full off-shell Einstein–Hilbert functional. If q_{coll} is identified with the already-represented Newtonian gravitational mode, its energy must *not* also be appended to Eq. (18), as this would double count the same physical static response. A separate dynamical q_{coll} field can be included only if its additional physical degrees of freedom and interactions are independently established.

What the $\hbar$-dependent force means. The massless-Dirac Boulware far-field result of Proposition 11.1 is obtained by varying the renormalized quantum matter functional on a classical effective metric, and solving the resulting semiclassical equations. At its stated asymptotic order, the supported-static-body Killing-energy potential has quantum term

$$\boxed{\delta\Phi_Q(r) = -\frac{2G_N^2 M\hbar}{15\pi c^3 r^3}, \qquad \delta F_r^{\text{coordinate}} = -m_t\,\partial_r\delta\Phi_Q = -\frac{2G_N^2 M m_t\hbar}{5\pi c^3 r^4}.} \tag{23}$$

Here $F_r^{\text{coordinate}}$ denotes the radial derivative of this specified Killing-energy potential; it is not automatically the locally measured proper holding force. The Boulware state, asymptotically flat source, massless four-component Dirac field and far-field limit are essential. This is the *same* one-loop quantum matter effect already represented in the metric solution, not a second independent $-\lambda\mathcal{Q}$ contribution. The two reconstructed exterior natural-invariant components need not be null at this quantum order (Eq. (65)). It is not a QGD-only prediction beyond semiclassical Einstein gravity.

Remark 3.8 (Scope of the word "final")**.** *Eq.* (18) *is the most complete* currently justified semiclassical effective action*, not a UV-complete quantum-gravity action. The signed-sector variables themselves are not quantized here. A genuine QGD-specific correction would require a newly justified independent functional* $\Delta\Gamma_{\text{sector}}$*; we set* $\Delta\Gamma_{\text{sector}} = 0$ *in the accepted action, rather than include an unspecified operator as if it were derived. The effective action's metric-only character preserves Chapter 4's quantum-equivalence result: the same renormalized matter content and state yield the same observables as semiclassical GR in the regime.*

4 Full master-metric variation and the QGD equivalent of Einstein's equations (QGD–137)

Scope and correction. The signed-sector identity $g = \bar{g} + \sum_s s_s\sigma^{(s)}\otimes\sigma^{(s)}$ is *only* the fixed-reference, $\lambda\mathcal{Q} = 0$ specialization of the founding master metric. It must never be promoted to the unrestricted QGD definition. In this section the reference coframe $T^a{}_\mu$ is fixed as a variational background (but transforms tensorially under a coordinate change), and the covectors are independent variables. The complete founding metric reads

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \mathcal{M}_{ab}, \quad \mathcal{M}_{ab} = \eta_{ab} - \sum_{s=1}^{N}\varepsilon_s\sigma_a^{(s)}\sigma_b^{(s)} - \lambda\mathcal{Q}_{ab}, \quad \varepsilon_s = \pm 1.} \tag{24}$$

The historical $\mathcal{Q}_{ab}$ is a *proposed* symmetric covariant correction, not a unique operator derived in Chapters 1–5. The different source-induced metric deformation q_{ab} discussed in Section 13 is a *solution-level* quantum backreaction defined through the semiclassical field equations. It is not to be inserted as a second independently varied term when the same renormalized matter functional already generates its effect.

4.1 The metric Euler tensor including quantum matter

For the renormalized metric-only action of Eq. (18), with its metric argument now interpreted as the full map (24) *when that map has been mathematically specified*, write

$$\mathcal{E}_g^{\mu\nu} := \frac{1}{\sqrt{-g}}\frac{\delta\Gamma^{\text{ren}}_{\text{metric}}}{\delta g_{\mu\nu}}, \qquad \Pi^{ab} := T^a{}_\mu T^b{}_\nu \mathcal{E}_g^{\mu\nu}. \tag{25}$$

Use the Chapter 4 stress convention $\delta S_{\text{matter}} = (1/2)\int\sqrt{-g}\,T^{\mu\nu}\delta g_{\mu\nu}$, and define the renormalized quantum source by

$$T_{\text{q}}^{\mu\nu} := \frac{2}{\sqrt{-g}}\,\frac{\delta(\hbar\Gamma^{(1)}_{\text{m,ren}} + O(\hbar^2))}{\delta g_{\mu\nu}}. \tag{26}$$

Within the chosen single-metric variational shorthand,

$$\boxed{\mathcal{E}_g^{\mu\nu} = -\frac{1}{16\pi G_N}\left[G^{\mu\nu}[g] + \Lambda_{\rm ren} g^{\mu\nu} - 8\pi G_N\left(T_{\rm cl}^{\mu\nu} + T_{\rm q}^{\mu\nu}\right)\right].} \tag{27}$$

The $T_{\rm q}$ notation covers matter-loop/state backreaction; for causal real-time evolution one must extract the physical equations from an in–in (closed-time-path) functional, not identify an in–out effective action with a causal time evolution equation.

4.2 The exact projected natural-invariant Euler equation

Theorem 4.1 (Full master-map stationarity (D|metric-only action))**.** *Let $\mathcal{Q}_{ab} = \mathcal{Q}_{ab}[\sigma; T]$ be a specified, functionally differentiable symmetric operator built from the covectors and fixed reference data; assume the metric is Lorentzian and boundary terms make the variation legitimate. Write $J_\sigma := D_\sigma g$ for the* complete *metric-map derivative. Stationarity in every independent sector is exactly*

$$\boxed{J_\sigma^*\left(\sqrt{-g}\,\mathcal{E}_g\right) = 0,} \tag{28}$$

or, in reference-frame notation with the volume-weighted adjoint,

$$\boxed{\mathscr{E}^{\,c}_{(s)} := \frac{1}{\sqrt{-g}}\frac{\delta\Gamma^{\rm ren}_{\rm metric}}{\delta\sigma_c^{(s)}} = -2\varepsilon_s\Pi^{cb}\sigma_b^{(s)} - \lambda\left[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\right]^c = 0.} \tag{29}$$

These are the full metric-pullback field equations *of QGD. If a separately justified $\Delta\Gamma_{\rm sector}[\sigma]$ exists, its independent Euler derivative must be added to the left-hand side and the metric-only equivalence below need not hold.*

Proof. With $\delta T^a{}_\mu = 0$,

$$\delta g_{\mu\nu} = T^a{}_\mu T^b{}_\nu\left[-\sum_s \varepsilon_s\left(\delta\sigma_a^{(s)}\sigma_b^{(s)} + \sigma_a^{(s)}\delta\sigma_b^{(s)}\right) - \lambda\delta\mathcal{Q}_{ab}\right]. \tag{30}$$

Substitute into $\delta\Gamma^{\rm ren}_{\rm metric} = \int\sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\delta g_{\mu\nu}$. Symmetry of Π^{ab} yields the algebraic contribution $-2\varepsilon_s\Pi^{cb}\sigma_b^{(s)}$; integrate the $D_s\mathcal{Q}$ contribution by parts, with admissible variations/boundary conditions, to obtain its formal adjoint. This proves Eqs. (28)–(29). □

For a local trial depending only on sectors and their first reference derivatives, write $\mathcal{Q}_{ab} = \mathcal{Q}_{ab}(\sigma, \partial\sigma; T)$. Define $C_{ab}^{(s)c} := \partial\mathcal{Q}_{ab}/\partial\sigma_c^{(s)}$ and $D_{ab}^{(s)\mu c} := \partial\mathcal{Q}_{ab}/\partial(\partial_\mu\sigma_c^{(s)})$. Then the explicit (coordinate) Euler equation is

$$\boxed{\begin{aligned} 0 = &-2\varepsilon_s\Pi^{cb}\sigma_b^{(s)} - \lambda\Pi^{ab}C_{ab}^{(s)c} \\ &+ \frac{\lambda}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\,\Pi^{ab}D_{ab}^{(s)\mu c}\right). \end{aligned}} \tag{31}$$

Derivatives of $\mathcal{Q}$ and the volume density are included, not silently set to zero. If instead $\mathcal{Q}$ uses the *dynamical* Levi–Civita derivative of g, then $D_g\mathcal{Q}$ also contributes to $\delta\mathcal{Q}$; Eq. (31) cannot be used unchanged. The full metric may then be implicit, and the appropriate full J_σ must first be determined.

4.3 When the projected equations are equivalent to EFE

Theorem 4.2 (Conditional quantum-corrected Einstein correspondence)**.** *Assume (i) the physical action depends on the covectors only through $g[\sigma]$, (ii) the full linearized metric map J_σ has a right inverse on the admissible symmetric metric variations with the chosen boundary conditions, and (iii) the renormalized matter variations and their state are treated consistently. Then*

$$\boxed{\mathscr{E}^{\,c}_{(s)} = 0 \quad \forall s, c \quad \Longleftrightarrow \quad G_{\mu\nu}[g] + \Lambda_{\mathrm{ren}} g_{\mu\nu} = 8\pi G_N \big(T^{\mathrm{cl}}_{\mu\nu} + T^{\mathrm{q}}_{\mu\nu}\big).} \tag{32}$$

In particular the equivalence is exact on Chapter 2's regular full-rank signed-sector branch at $\lambda = 0$. This theorem does not assert that the right-inverse property holds for an arbitrary, nonzero historical $\lambda\mathcal{Q}$.

Proof. If $\mathcal{E}_g = 0$, then $J_\sigma^* \mathcal{E}_g = 0$ by Eq. (28). Conversely, for any admissible δg, choose $\delta\sigma$ with $J_\sigma \delta\sigma = \delta g$. Vanishing of the sector variation implies $\int \sqrt{-g}\, \mathcal{E}_g^{\mu\nu} \delta g_{\mu\nu} = 0$ for every symmetric variation, hence $\mathcal{E}_g = 0$ and Eq. (32) follows from Eq. (27). □

Proposition 4.3 (Stability of the full-rank branch under a controlled correction)**.** *Let the linearized map act between fixed Banach spaces of admissible sector and metric perturbations. Suppose the $\lambda = 0$ map J_0 admits a* bounded *right inverse R_0, $J_0 R_0 = I$, and $J_\lambda = J_0 + \lambda B$ on these spaces, where BR_0 is bounded and $\|\lambda B R_0\| < 1$. Then J_λ also has a bounded right inverse, namely*

$$\boxed{R_\lambda = R_0 \,(I + \lambda B R_0)^{-1}, \qquad J_\lambda R_\lambda = I.} \tag{33}$$

Consequently the full-map EFE correspondence survives every specified *correction satisfying these regularity and smallness conditions. These conditions are not verified for an unspecified historical $\mathcal{Q}$.*

Proof. $J_\lambda R_0 = I + \lambda B R_0$; the Neumann series converges for $\|\lambda B R_0\| < 1$. Multiplication by its inverse gives Eq. (33). □

4.4 Classical limit, the quantum metric term, and the collective bridge

If the chosen completion has $\lambda\mathcal{Q} \to 0$ as $\hbar \to 0$, loop terms vanish at fixed measured G_N, Λ and smooth classical sources, and the metric Jacobian remains regular, Eq. (32) reduces to $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T^{\mathrm{cl}}_{\mu\nu}$. For finite $\hbar$, the same metric-only action gives the quantum-stress source in Eq. (32); solving for g and decomposing it afterward as $g = g_{\mathrm{cl}} + q$ does not require adding a second independent q kinetic term. Chapter 5's $G_N = g_c^2/(4\pi Z)$ remains a *static matching condition* on the measured coupling, not a new metric operator. The per-wave $\widetilde{g}^{(\psi)} = \Omega_\psi^2 g$ of Section 13 remains an effective phase-trajectory description for a specified probe, not the universal spacetime metric inserted into the gravitational action. If the reference coframe is itself promoted to a dynamical field, its variation yields *additional* Euler equations; this section does not silently vary T.

4.5 Degrees of freedom of the QGD action (QGD–121)

S_{QGD} is written in terms of $4N$ sector components, more than the ten components of g. Around a background $\bar\sigma$, the action depends on $\delta\sigma$ only through $\delta g = D(\bar\sigma)\,\delta\sigma$, so its quadratic kernel is

$$K_\sigma(k) = D^{\mathsf{T}} K_{\mathrm{FP}}(k)\, D, \tag{34}$$

with K_{FP} the linearized Einstein–Hilbert kernel. Gauge directions are the null vectors of K_σ at generic k. Propagating modes are the additional null vectors that appear on the light cone, and dynamical components are those with a k_0^2 term.

Theorem 4.4 (Degrees of freedom (D, N))**.** *On any spanning branch (*$\operatorname{rank} D = 10$*):*

1. *the* $4N-10$ *directions of* $\ker D$ *are the tangent space of the local sector-rotation orbit (*$O(1,3)$ *for four* η*-signed sectors,* $O(4,4)$ *for eight). Since this equals the dimension of the fibre of* $\sigma \mapsto g$*, the fibres are locally orbits, and the redundancy is an exact local gauge symmetry of the nonlinear action;*

2. *together with the four diffeomorphism directions these exhaust the gauge null space; the* k_0^2 *part of* K_σ *has rank* 6*, as in general relativity;*

3. *exactly* ***two*** *modes propagate:* $4N - (4N-10) - 4 - 4 = 2$*.*

Background	comps	$\ker D$	null, generic k	null, $k^2 = 0$	propagating	k_0^2 rank
GR (reference)	10	–	4	6	2	6
4 sectors, $\sigma^{(a)} = v\,\mathrm{d}x^a$	16	6	10	12	2	6
8 sectors, cancelling background	32	22	26	28	2	6
8 sectors, random background	32	22	26	28	2	6
$\sigma = 0$	16	16	16	16	0	0
1 sector, timelike	4	0	2	2	0	0
1 sector, null	4	0	2	2	0	2

Verification. Ranks computed numerically from Eq. (34); GR's own count (4, 6, 2) reproduced as a control. At random spanning backgrounds the group orbit has dimension 6 (four sectors) and 22 (eight sectors), equal to $\dim\ker D$, with every orbit direction in the kernel to 10^{-16}. Random backgrounds and random momenta always give two propagating modes.

Proposition 4.5 (Degenerate and non-spanning branches (D, N))**.** *1. At* $\sigma = 0$*,* $D = 0$ *and the quadratic action vanishes identically. The metric there is Minkowski, but* $\sigma = 0$ *is a degenerate point of the variables: perturbation theory about it has no kinetic terms. The vacuum must be represented on a spanning branch, such as* $\sigma^{(a)} = v\,\mathrm{d}x^a$ *with signs* $(-1,+1,+1,+1)$ *or the eight-sector cancelling background, which describe the same metric with non-degenerate variables.*

2. *On one-sector branches nothing propagates at linear order: the timelike branch has no time-derivative terms, and the null (Kerr–Schild-type) branch has two but no light-cone modes. The sector equations there are projections of Einstein's equations (Chapter 2) and do not reproduce the radiative sector of general relativity, consistent with Chapter 2's pp-wave counterexample.*

Consequence. On spanning branches the extra sector components are pure gauge, there are no additional or ghost modes, and the QGD action has exactly the dynamical content of general relativity. The spanning condition, previously a requirement for the equivalence of the field equations, is also the condition for the action to have the right degrees of freedom.

Proposition 4.6 (Spanning representations of the chapters' backgrounds (QGD–122))**.** *1.* ***Kerr–Schild class.*** *For any metric* $g = \eta + H\,k \otimes k$ *with* k *null (Schwarzschild, Reissner–Nordström, Vaidya, the regular cores, the pp-waves, Kerr and Kerr–Newman), the four-sector lift* $\Sigma = v\mathbb{1} + \frac{H}{2v}(\eta k)\,k^{\mathsf{T}}$ *with signs* $(-1,+1,+1,+1)$ *and flat reference* $(1-v^2)\eta$ *(*$0 < v < 1$*) reproduces* g *exactly, and* $\det\Sigma = v^4$ *because* k *is null. It is therefore spanning wherever* H *is finite, including at and inside horizons.*

2. ***Cosmology.*** *Chapter 2's conformal-time FLRW lift (pairs in all four directions) is spanning. Its cosmic-time catalogue lift (spatial pairs only), and the same construction for Kasner, has rank* 9*: the lapse direction* g_{00} *is unreachable. Adding one cancelling time pair,* $P_0 = N_0 = v$*, makes it spanning without changing the metric.*

3. ***One-sector reductions.*** *For $\sigma = A\,\mathrm{d}t + B\,\mathrm{d}r$ the reduced equations are exactly two projections of the Einstein tensor,*

$$\mathcal{E}_A = -\sqrt{-g}\,(2AG^{tt} + 2BG^{tr}), \qquad \mathcal{E}_B = -\sqrt{-g}\,(2AG^{tr} + 2BG^{rr}). \tag{35}$$

Within static spherical symmetry they determine the solution (symmetric criticality): at their solution $A^2 = B^2 = C/r$ every component of $G^{\mu\nu}$ vanishes, and with matter they reproduce the TOV benchmarks. They do not govern perturbations outside the symmetry class.

Verification. Item 1: Schwarzschild and Kerr ($a = 0.9M$) at $r/M = 10$, both horizons, between them, and $r/M = 0.05$; $\det\Sigma/v^4 = 1.00000000$, rank 10, and metric reproduced to round-off throughout. Item 2: rank 9 with g_{00} the unique unreachable direction for the spatial lift; rank 10 for the conformal-time lift ($a = 10^{-3}$–3) and for the augmented lift ($a = 10^{-3}$–50). Item 3: both ratios in Eq. (35) computed symbolically as exactly -1, and all ten components of $G^{\mu\nu}$ vanish symbolically at the reduced solution.
Consequence. Every background used in Chapters 2–4 has a spanning QGD representation, so by Theorem 4.4 perturbations about black holes and cosmologies carry exactly the dynamics of general relativity. The one-sector and spatial-only representations remain valid for the backgrounds themselves but must be replaced by their spanning lifts for perturbation theory.

5 What changes in the unrestricted derivative-dependent metric map

The master proposal (6) permits a nonzero term $-\lambda\mathcal{Q}_{ab}$. Its functional form and coefficient are currently unmatched; the existence of a symbol $\mathcal{Q}_{ab}$ does not establish the independent field equation $\Box_g\sigma_\mu$ or any particular force correction. A conditional chain rule can, however, be proved *before* choosing that operator.

Proposition 5.1 (Functional chain rule for a first-derivative metric map)**.** *Hold the reference coframe fixed and suppose the composite metric is a smooth local function $g_{\mu\nu}(\sigma, \partial\sigma)$ of the independent natural-invariant fields and their first coordinate derivatives. Assume the matter and gravitational action depend on σ only through the resulting metric, and its metric Euler variation is $\delta\Gamma = \int \mathrm{d}^4x\,\sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\,\delta g_{\mu\nu}$ after the appropriate boundary treatment. Then, for arbitrary compactly supported $\delta\sigma$, the natural-invariant Euler derivative obeys*

$$\boxed{\frac{\delta\Gamma}{\delta\sigma_\rho^{(s)}} = \sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\frac{\partial g_{\mu\nu}}{\partial\sigma_\rho^{(s)}} - \partial_\alpha\left(\sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\frac{\partial g_{\mu\nu}}{\partial(\partial_\alpha\sigma_\rho^{(s)})}\right).} \tag{36}$$

If g depends on second or higher derivatives of σ, the corresponding alternating higher total derivatives are required. Direct nonmetric σ couplings add their own *Euler functional derivatives.*

Proof. Use the ordinary chain rule on the specified local metric map, substitute into the metric Euler variation, and integrate by parts the term proportional to $\partial_\alpha\delta\sigma_\rho^{(s)}$. Compact support removes the boundary contribution. A second integration-by-parts argument handles second derivatives, and so on. □

This equation expresses *how* a well-defined derivative metric operator would enter; it does not determine an independent kinetic coefficient from the already accepted action. Without the actual $\mathcal{Q}_{ab}$ and its admissible boundary terms, no numerical quantum-force coefficient can be assigned to this contribution.

Remark 5.2 (The derivative channel on Chapter 2's constructed branch (QGD–101/105))**.** *For both trial completions of $\mathcal{Q}$ in Chapter 2 and any $\lambda \neq 0$, Chapter 2 constructs a particular nine-sector branch on which the metric map has a differential right inverse (Theorems on the nine-sector lift). On that branch, a metric-mediated action gives exactly the effective metric equations, whatever λ is. As Chapter 2 states, this concerns a particular spanning branch, not all field configurations; on other branches, or with σ dependence not passing through g, the question remains open.*

6 Does this equation say anything beyond GR?

One projected field equation per covector need not contain all ten independent symmetric metric equations. What matters is the rank of the map from independently variable signed covectors to the ten metric components.

Proposition 6.1 (Exact regular signed section (D))**.** *In a fixed local reference frame let $h_{\mu\nu} = g_{\mu\nu} - \bar{g}_{\mu\nu}$ be any real symmetric 4×4 matrix. For any constant $a > 0$ set*

$$\boxed{P = aI_4 + \frac{h}{4a}, \qquad N = aI_4 - \frac{h}{4a}.} \tag{37}$$

The four rows of P and four rows of N define eight signed covectors, and

$$\boxed{P^{\mathsf{T}}P - N^{\mathsf{T}}N = h.} \tag{38}$$

For every symmetric variation δh, the choice $\delta P = \delta h/(4a)$ and $\delta N = -\delta h/(4a)$ yields $\delta g = \delta h$. Thus the metric differential has rank ten on this section.

Proof. Expand both products in (38); the $a^2 I$ and $h^2/(16a^2)$ terms cancel, leaving h. Differentiation gives the displayed right inverse. □

Theorem 6.2 (Conditional equivalence with Einsteinian dynamics (D|A))**.** *Suppose the full effective action depends on the signed covectors* only through $g(\sigma)$*, and the admissible covector variations possess a local right inverse to arbitrary symmetric metric variations, as on the section* (37)*. Then*

$$\boxed{\mathscr{S}^{\rho}_{(s)} = 0 \text{ for every sector } s \quad \Longleftrightarrow \quad \mathcal{E}_g^{\mu\nu} = 0.} \tag{39}$$

Thus, on this regular full-rank branch, the action-derived natural-invariant equations are equivalent to the specified *effective Einstein–matter equations.*

Proof. If $\mathcal{E}_g = 0$, every projected Euler derivative is zero. Conversely, let every covector Euler derivative vanish. Use the differential right inverse of Proposition 6.1 to express any compactly supported symmetric metric test variation as an admissible signed-covector variation. Then $\delta\Gamma = 0$ for all metric tests and the fundamental lemma of the calculus of variations gives $\mathcal{E}_g = 0$. □

Interpretation. Writing the Einsteinian equations using natural invariants is a nontrivial local representation and an alternative choice of variables. It does *not*, by itself, generate additional observables beyond GR. In the metric-only sector the transformation $S \mapsto O(x)S$ with $S = \binom{P}{N}$ and $O^{\mathsf{T}}\mathrm{diag}(I_4, -I_4)O = \mathrm{diag}(I_4, -I_4)$ leaves g unchanged. The resulting local $O(4,4)$ freedom is a parametrization redundancy wherever it is a symmetry of the complete action and matter couplings. A physical σ representative need not be unique; adding direct interactions could change this symmetry.

7 The calibrated gravitational field and the normalization distinction

Chapter 3 independently varies both components $A(r)$ and $B(r)$ of the declared classical spherical one-form $\sigma = A\,\mathrm{d}x^0 + B\,\mathrm{d}r$, without setting $A = B$ in the action. On its nondegenerate, asymptotically flat vacuum branch, the adopted Einstein–Hilbert sector gives

$$\boxed{A(r)^2 = B(r)^2 = \frac{C_g}{r}, \qquad C_g = \frac{2G_N M}{c^2} =: r_s.} \tag{40}$$

The value of C_g is fixed *once* by the Newtonian exterior acceleration. It is not derived from the microscopic Dirac equation; its calibration is legitimate correspondence with the observed Newtonian mass parameter, as in the historical Einstein normalization. No other coefficient is adjusted to recover higher orders.

When an independently selected reference fermion mass $m_{\mathrm{ref}} > 0$ supplies a reduced Compton length $\bar{\lambda}_{\mathrm{ref}} = \hbar/(m_{\mathrm{ref}}c)$, Chapter 3 defines the dimensionless *geometric inverse-strength convention*

$$\boxed{|\alpha_G^{\mathrm{geom}}|^2 := \frac{\bar{\lambda}_{\mathrm{ref}}}{C_g} = \frac{\hbar c}{2G_N M m_{\mathrm{ref}}}.} \tag{41}$$

The original common spinor multiplier α_G^{spin} in $\Psi = (\alpha_G^{\mathrm{spin}}/r)\sum_a \psi_a u_a e^{\pm iS_a/\hbar}$ is *not* thereby normalized from the geometric equations. Indeed, the simultaneous change $\alpha_G^{\mathrm{spin}} \mapsto z\alpha_G^{\mathrm{spin}}$, $\psi_a \mapsto \psi_a/z$ with $z \neq 0$ leaves the physical state unchanged. Its absolute normalization comes from an independently specified current or occupied-state convention. In particular, $\alpha_G^{\mathrm{spin}} = \alpha_G^{\mathrm{geom}}$ is not a theorem of Chapters 1–3.

The observable static Killing-energy potential per unit held test mass is

$$\boxed{\Phi_{\mathrm{stat}}(r) = c^2\left(\sqrt{1 - \frac{C_g}{r}} - 1\right), \qquad r > C_g.} \tag{42}$$

Its convergent, wholly *classical* expansion is

$$\boxed{\Phi_{\mathrm{stat}}(r) = -\frac{G_N M}{r} - \frac{G_N^2 M^2}{2c^2 r^2} - \frac{G_N^3 M^3}{2c^4 r^3} - \frac{5G_N^4 M^4}{8c^6 r^4} + \cdots.} \tag{43}$$

All powers of $\hbar$ cancel when (41) is used merely as a parametrization of C_g. Thus these higher orders are not evidence of quantum-corrected vacuum gravity.

8 The microscopic Dirac bridge: what it fixes

For one locally on-shell, positive-energy, outward-propagating Dirac WKB branch with conserved particle flux $\dot{N} > 0$, Chapter 1 and Chapter 3 give the magnitude-only current identity

$$\boxed{|\alpha_G^{\mathrm{spin}}\psi_0(r)|^2 = \frac{\dot{N}\,\Delta(r)}{8\pi c^2 p(r)}, \qquad \Delta = E_{\mathrm{loc}} + m_{\mathrm{eff}}c^2, \quad p > 0.} \tag{44}$$

This is a transport constraint. It determines the envelope only relative to an independently fixed momentum profile and flux; it does *not* determine a $1/r$ gravitational potential, the geometric normalization, or an absolute phase. A static bound source with zero net radial current is *not* normalized using this outgoing flux formula. Its spinor profiles must be obtained from its own coupled boundary-value equations with conserved occupation.

9 A controlled quantum matter contribution to the source

To make the $\hbar$ dependence precise without silently inserting a new gravitational operator, use a stated positive-energy, fixed-spin, nonrelativistic Dirac-to-Pauli limit and write $\varphi = \sqrt{n}\, e^{iS/\hbar}\chi$ with constant unit Pauli spinor χ, $n > 0$, and fermion mass $m_f > 0$. The polar identity for the Schrödinger kinetic energy is

$$\boxed{E_{\rm kin}[n,S] = \int \mathrm{d}^3x \left[\frac{n|\boldsymbol{\nabla} S|^2}{2m_f} + \frac{\hbar^2}{8m_f}\frac{|\boldsymbol{\nabla} n|^2}{n}\right].} \tag{45}$$

The quantum density-gradient functional and associated momentum-flux tensor (one admissible improvement convention) satisfy

$$\boxed{Q[n] = -\frac{\hbar^2}{2m_f}\frac{\nabla^2\sqrt{n}}{\sqrt{n}}, \qquad \Pi^Q_{ij} = \frac{\hbar^2}{4m_f}\left(\frac{\partial_i n \partial_j n}{n} - \partial_i\partial_j n\right), \qquad \partial_j \Pi^Q_{ij} = n\partial_i Q.} \tag{46}$$

This is a derived coefficient of *quantum matter stress*, not a derived coefficient of a standalone $\hbar^2(\nabla\sigma)^2$ gravitational kinetic term. To leading nonrelativistic order, the *adopted* sourced Einstein–Hilbert equations reduce to Poisson's law:

$$\boxed{\nabla^2\Phi = 4\pi G_N m_f n, \qquad \mathcal{U} := -\frac{2\Phi}{c^2}, \qquad \nabla^2\mathcal{U} = -\frac{8\pi G_N m_f}{c^2} n.} \tag{47}$$

If the physical source profile $n(x;\hbar)$ is independently determined from its quantum dynamics, its $\hbar$ dependence influences the sourced metric and consequently the corresponding temporal natural-invariant *combination*. This is a source effect, not a new vacuum propagation law. Relativistic stress, pressure, spin, multi-orbital occupations and binding require the complete Dirac–Einstein source instead of simply substituting one n.

Proposition 9.1 (Compact spherical exterior at fixed mass (D|A))**.** *Let $\rho_\hbar(r)$ be an integrable, nonnegative spherically symmetric mass density with fixed total mass $M = 4\pi\int_0^\infty \rho_\hbar(r) r^2\,\mathrm{d}r$, compactly supported in $r \le R$. Assume Poisson's equation and $\Phi(\infty) = 0$. Then $\Phi(r) = -G_N M/r$ and $F_r = -G_N M m_t/r^2$ for every $r > R$, independently of the interior $\hbar$ dependence.*

Proof. Spherical Poisson integration gives $r^2\Phi'(r) = 4\pi G_N \int_0^r \rho_\hbar(s) s^2\,\mathrm{d}s$. The integral equals $M/(4\pi)$ outside the support. Integrate once more with the stated asymptotic boundary condition. □

Thus a universal exterior term proportional to $G_N M\hbar^2/(m_f^2 c^2 r^4)$ does *not* follow from changing only the profile of a compact spherical quantum source while keeping its mass and the Poisson law fixed. Noncompact sources, altered total mass, non-spherical multipoles or independently derived quantum field equations are distinct questions.

Remark 9.2 (Varying spin (QGD–113))**.** *The fixed-spin assumption above is a special case. When the spin direction $\hat{s}$ varies, the exact spin-$\frac{1}{2}$ flow law of Section 13.5 adds the spin quantum potential $(\hbar^2/8m_f)|\nabla\hat{s}|^2$ and a spin-vorticity force; both vanish for constant $\hat{s}$, recovering Eq. (46).*

10 Status of the historical quantum-force candidate

The proposed universal attractive force

$$\boxed{F_{r,\rm hist} = -\frac{G_N M m_t}{r^2}\left[1 + \frac{9}{2}\left(\frac{\hbar}{m_f c r}\right)^2 + \cdots\right]} \tag{48}$$

remains a historical *candidate*, not a conclusion of the accepted action. The reciprocal of $e^{2im_fcr/\hbar}$ is $e^{-2im_fcr/\hbar}$: inverting nonconstant Taylor terms separately does not invert the complete exponential. The historical cubic energy also gives a sign opposite to an enhancement of attraction.

A first-order weak-field expansion of a specified Liouville-normal, separated Schwarzschild radial Dirac equation produces coefficients depending on its angular channel κ. Their derivatives are not coordinate-, tetrad-, or field-redefinition-invariant force observables. In particular the lowest $\kappa = -1$ channel does not produce the historical term at order $G_N\hbar^2/r^4$ in that normal form. Nothing here establishes a universal quantum change to the exterior gravitational acceleration. To claim one, first derive a covariant $\hbar$-dependent action, vary it including all connection and metric dependence, and calculate a gauge-/representation-invariant physical observable.

10.1 Independent referee test of the radial Dirac coefficient (QGD–098)

The following is a *conditional theorem about a quantum test field.* It does not modify the natural-invariant field equations. Adopt the classical, Newton-calibrated Schwarzschild exterior of Eq. (40), minimally couple a massive four-component Dirac field through the induced orthonormal-frame spin connection, and use the standard separated, fixed-tetrad radial basis. This additional coupling and the reduction to a single radial mode are assumptions of the *test-field model*, not consequences of the flat-space spinor-component ratios alone. Keep $r > r_s > 0$ and define

$$\ell_C := \frac{\hbar}{m_f c}, \quad x := \frac{r}{\ell_C}, \quad a := \frac{r_s}{\ell_C}, \quad e := \frac{E}{m_f c^2}, \quad F := 1 - \frac{a}{x}, \quad \frac{\mathrm{d}}{\mathrm{d}x_*} := F\frac{\mathrm{d}}{\mathrm{d}x}, \tag{49}$$

where $\kappa \in \mathbb{Z} \setminus \{0\}$ is the spinor angular index. For the stated radial spinor gauge the first-order field equations read

$$\boxed{\begin{aligned} \frac{\mathrm{d}f}{\mathrm{d}x_*} &= (e + \sqrt{F})g - \frac{\kappa\sqrt{F}}{x}f, \\ \frac{\mathrm{d}g}{\mathrm{d}x_*} &= \frac{\kappa\sqrt{F}}{x}g + (\sqrt{F} - e)f. \end{aligned}} \tag{50}$$

Proposition 10.1 (Liouville-normal radial coefficient (D|A))**.** *Assume Eqs. (49)–(50), $e = 1$, and $a/x < 1$. Put $B := 1 + \sqrt{F}$, $K := \kappa\sqrt{F}/x$, and*

$$P := \frac{F_x}{F} - \frac{B_x}{B}, \qquad Q := FK_x - F\frac{B_x}{B}K - K^2 + 1 - F. \tag{51}$$

Then elimination of g gives

$$f_{xx} + Pf_x + \frac{Q}{F^2}f = 0. \tag{52}$$

After the invertible local Liouville rescaling $f = \exp(-\frac{1}{2}\int P\,\mathrm{d}x)u$, the transformed equation is

$$\boxed{u_{xx} + \mathcal{V}_\kappa(x,a)u = 0, \qquad \mathcal{V}_\kappa = \frac{Q}{F^2} - \frac{1}{2}P_x - \frac{1}{4}P^2.} \tag{53}$$

Its expansion at fixed $x > 0$ about $a = 0$ is

$$\boxed{\begin{aligned} \mathcal{V}_\kappa = &-\frac{\kappa(\kappa+1)}{x^2} + \frac{a}{x} + \frac{3 - 4\kappa^2 - \kappa}{4}\frac{a}{x^3} \\ &+ \frac{2a^2}{x^2} + \frac{-64\kappa^2 + 4\kappa + 69}{64}\frac{a^2}{x^4} + O(a^3). \end{aligned}} \tag{54}$$

In particular,

$$\boxed{\mathcal{V}_{-1}(x,a) = \frac{a}{x} + \frac{2a^2}{x^2} + \frac{a^2}{64x^4} + O(a^3),} \tag{55}$$

so the $O(a/x^3)$ correction vanishes in the $\kappa = -1$ channel. The displayed expansions are local analytic expansions under the stated hypotheses; they are not uniform near $r = r_s$.

Proof. Solve the first equation in (50) for $g = (Ff_x + Kf)/B$. Differentiate with respect to x_*, substitute the second equation, and collect the f_x and f coefficients to obtain Eq. (52). Direct differentiation of the Liouville rescaling gives Eq. (53). Using $\sqrt{1-a/x} = 1 - a/(2x) - a^2/(8x^2) + O(a^3)$ in the resulting *exact* coefficient gives Eq. (54). Setting $\kappa = -1$ proves Eq. (55). □

Remark 10.2 (What is and is not a quantum force)**.** *At $e = 1$ one may* formally *rewrite Eq. (53) as a zero-energy one-dimensional Schrödinger equation, with $U_{\text{radial}} := -m_f c^2 \mathcal{V}_\kappa/2$. Its $O(a/x^3)$ contribution is*

$$\delta U_\kappa = -\gamma_\kappa \frac{G_N M \hbar^2}{m_f c^2 r^3}, \qquad \gamma_\kappa := \frac{3 - 4\kappa^2 - \kappa}{4}. \tag{56}$$

Differentiating this representation-dependent radial term *yields $-3\gamma_\kappa G_N M \hbar^2/(m_f c^2 r^4)$, but this is* ***not*** *by itself a local force observable or an invariant correction to the gravitational metric. In particular, $\gamma_{-1} = 0$ and $\gamma_{+1} = -1/2$. The historical* attractive *coefficient $9/2$ would require $\gamma_\kappa = 3/2$, i.e. $4\kappa^2 + \kappa + 3 = 0$, which has no real root. The $a^2/(64x^4)$ term at $\kappa = -1$ is likewise an* effective radial-equation coefficient*, not a new vacuum Einstein tensor. It must not be reported as a distinct QGD quantum-gravitational force.*

Referee/proof boundary. The result above rests on a stated classical Schwarzschild exterior, the minimally coupled test Dirac equation, a fixed tetrad and spinor angular basis, and a specified threshold energy $e = 1$. It therefore follows from Chapters 1–2 only *after* their declared effective metric/Dirac coupling is adopted. It is not derived solely from the flat-space macroscopic spinor ratios. Exact symbolic differentiation verifies Eqs. (52)–(55); this is not a Lean-kernel proof. A physical quantum-gravity claim would additionally require the modified, conserved metric/source field equations and an invariant observable, with no further fit to the historical $9/2$ coefficient.

11 A published semiclassical Dirac benchmark in natural-invariant variables

The historical $9/2$ Compton-scale force is not established by the original phase manipulation. There is, however, a distinct and quantitatively fixed $\hbar$-dependent *gravitational* correction that can be imported into the effective metric sector. Carlson, Anderson, Fabbri, Fagnocchi, Hirsch and Klyap computed the leading far-field renormalized stress tensor of one free *massless four-component* Dirac field in an asymptotically empty (Boulware) state, and checked its gravitational backreaction against previous one-loop calculations [7]. This is a benchmark of semiclassical Einstein–Dirac physics, not an independent derivation of gravitational dynamics from the original flat-space QGD Dirac action.

Proposition 11.1 (Massless Dirac far-field gravitational correction (D|A))**.** *Adopt the metric-only effective Einstein–Hilbert action with a minimally coupled quantized free massless four-component Dirac field, the renormalized Boulware state, asymptotic flatness, fixed ADM mass $M > 0$, areal radius r, and leading-order semiclassical backreaction. In units $G_N = c = \hbar = 1$, the published leading stress tensor is*

$$\boxed{\langle T^{\mu}{}_{\nu} \rangle_{\text{ren}} = \frac{M}{40\pi^2 r^5}\, \text{diag}(4, 2, -3, -3) + O(r^{-6}).} \tag{57}$$

For the far-field static metric

$$\mathrm{d}s^2 = -f(r)\mathrm{d}t^2 + \left(1 - \frac{2\mathfrak{m}(r)}{r}\right)^{-1} \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2, \qquad f(r) = \left(1 - \frac{2\mathfrak{m}(r)}{r}\right) e^{2\nu(r)}, \tag{58}$$

the asymptotically normalized functions satisfy

$$\mathfrak{m}(r) = M\left(1+\frac{1}{5\pi r^2}\right)+O(r^{-3}), \tag{59}$$

$$\nu(r) = \frac{M}{15\pi r^3}+O(r^{-4}), \tag{60}$$

$$f(r) = 1-\frac{2M}{r}-\frac{4M}{15\pi r^3}+O(r^{-4}), \tag{61}$$

where in the last expression $O(r^{-4})$ also includes mixed classical–semiclassical orders. Restoring $G_N, c, \hbar$, the leading far-field *supported-static-body Killing-energy potential per unit test mass is*

$$\boxed{\begin{aligned}\Phi^{\mathrm{sc}}_{\mathrm{QGD}}(r) = c^2\left(\sqrt{1-\frac{2G_NM}{c^2r}}-1\right)\\ -\frac{2G_N^2M\hbar}{15\pi c^3r^3}+O\!\left(\frac{G_N^3M^2\hbar}{c^5r^4}\right)+O(\hbar^2),\end{aligned}} \tag{62}$$

where the final $O(\hbar^2)$ denotes omitted higher quantum orders, not a uniform SI-dimensional bound. The displayed quantum coefficient is meaningful in the far field $r \gg r_s = 2G_NM/c^2$, where the quantum backreaction is perturbatively small. It must not be extrapolated unchanged to $r \simeq r_s$.

Proof. For the ansatz (58), the radial Einstein equations at leading quantum order read

$$\mathfrak{m}' = -4\pi r^2\langle T^t{}_t\rangle_{\mathrm{ren}}, \qquad \nu' = \frac{4\pi r(\langle T^r{}_r\rangle_{\mathrm{ren}}-\langle T^t{}_t\rangle_{\mathrm{ren}})}{1-2\mathfrak{m}/r}. \tag{63}$$

Substituting (57) gives $\mathfrak{m}' = -2M/(5\pi r^3)$ and $\nu' = -M/(5\pi r^4)+O(r^{-5})$. Integrating with $\mathfrak{m}(\infty)=M$ and $\nu(\infty)=0$ proves (59)–(60). Consequently $f = 1-2M/r-4M/(15\pi r^3)+O(r^{-4})$. For an observer held static in this exterior, $\Phi = c^2(\sqrt{f}-1)$; expanding to first order in the renormalized Dirac stress and restoring dimensions proves (62). This is a sourced asymptotic expansion, not the solution of a freely chosen scalar force law. □

Remark 11.2 (Quantum gravitational force and Planck-scale residue)**.** *The* quantum part *of the signed radial force on a held static test mass m_t has leading far-field behavior*

$$\boxed{\delta F_r^{\mathrm{sc}} = -\frac{2G_N^2Mm_t\hbar}{5\pi c^3r^4}+\textit{higher orders.}} \tag{64}$$

Relative to the Newtonian attraction, its leading magnitude is $\frac{2}{5\pi}\frac{G_N\hbar}{c^3r^2}$, which is governed by the Planck length squared, not the inverse square of an arbitrarily selected probe mass. It is therefore not *the historical $(9/2)(\hbar/(m_fcr))^2$ correction. The classical piece $-m_t\Phi'_{\mathrm{stat}}$ remains defined by the exact background square root. The quantum term is a controlled far-field correction, not an exact all-radius force.*

Remark 11.3 (Natural-invariant reconstruction and status)**.** *The full signed/tetrad QGD master metric (6) is unchanged. For the nondegenerate one-sector static spherical* metric representation *discussed in Chapter 2, write $\sigma = A(r)\mathrm{d}t + B(r)\mathrm{d}r$ and put $F = 1-A^2$, $D = 1+B^2-A^2$ so that the diagonal radial metric coefficient is D/F. Reconstruct the semiclassical metric by $A^2 = 1-f$ and $D/F = (1-2\mathfrak{m}/r)^{-1}$. Then, in the far field and to the first displayed quantum order,*

$$\boxed{\begin{aligned}A^2(r) &= \frac{2M}{r}+\frac{4M}{15\pi r^3}+O(r^{-4}),\\ B^2(r) &= \frac{2M}{r}+\frac{2M}{5\pi r^3}+O(r^{-4})\end{aligned}} \qquad (G_N=c=\hbar=1). \tag{65}$$

Thus the familiar equality $A^2 = B^2$ is a property of the classical Schwarzschild branch, not *of the quantum-corrected source. This construction shows that QGD variables can represent the published semiclassical metric; it does not establish a new dynamical law for σ_μ.*

Assumptions and formal-verification boundary. The algebraic implications from (57) and (63) to (62) are suitable for a Lean-style theorem once analytic asymptotic bounds and positivity conditions on the metric are specified. The renormalized-stress coefficient in (57) is a *published quantum field-theoretic input*, not a theorem derived from Chapter 1. The complete original four-mode wavefunction has not been proved to generate this Boulware vacuum as its unique quantum state. At the current metric-only effective level, the equivalence theorem in the next section applies: this correction is already predicted by semiclassical GR with the same Dirac matter.

12 Can QGD predict a quantum force independently of semiclassical GR?

This is the central discriminating question of QGD–096. Here *semiclassical Einstein gravity* means a classical metric sourced by a renormalized expectation value or effective action of quantum matter; *quantized gravitational effective field theory* includes fluctuations of the gravitational field itself. The two comparisons are different. A Dirac wave in a prescribed metric, the quantum-potential term of a Pauli reduction, and an $\hbar$-dependent source profile are not, by themselves, quantum corrections to the gravitational vacuum response. The relevant test is an invariant observable calculated from an effective dynamics which QGD derives but which the selected GR-plus- quantum-matter comparator does not already possess.

Theorem 12.1 (Metric-only quantum-equivalence obstruction (D|A))**.** *Let $\mathcal{F}$ be the algebraic signed-sector map $\sigma \mapsto g[\sigma]$ on a regular branch whose differential is surjective onto symmetric metric variations, with identical boundary conditions and matter states on both sides. Let $\Gamma_{\rm sc}[g,\Psi]$ be a specified semiclassical Einstein–matter effective functional, possibly including renormalized quantum matter loops, and assume*

$$\boxed{\Gamma_{\rm QGD}[\sigma,\Psi] = \Gamma_{\rm sc}[\mathcal{F}(\sigma),\Psi]} \tag{66}$$

with no independent σ interaction or separately quantized σ measure. Then stationary QGD fields on this branch and stationary fields of $\Gamma_{\rm sc}$ determine the same set of physical metrics and metric/matter observables. In particular, no additional universal $\hbar$-dependent gravitational-force coefficient is implied solely by expressing g through σ.

Proof. For any admissible variation, the functional chain rule gives

$$\delta\Gamma_{\rm QGD} = \int \mathrm{d}^4x\,\sqrt{-g}\,\mathcal{E}^{\mu\nu}_{\rm sc}\,\frac{\partial g_{\mu\nu}}{\partial\sigma^{(s)}_\rho}\delta\sigma^{(s)}_\rho + \delta_\Psi\Gamma_{\rm sc}, \tag{67}$$

where repeated sector indices are summed. A stationary metric of $\Gamma_{\rm sc}$ therefore makes the QGD pullback stationary. Conversely, by surjectivity every compactly supported symmetric δg is induced by an admissible $\delta\sigma$; stationarity of the pullback and the independently imposed matter equations imply $\mathcal{E}^{\mu\nu}_{\rm sc} = 0$. Because all specified observables are functionals of the same physical metric and matter state, they agree. The result is an on-shell, mean-effective-action theorem: it makes no claim about quantizing the redundant variables with a new functional measure, about rank-deficient branches, or about a new action not covered by (66). □

Remark 12.2 (What the theorem does, and does not, rule out)**.** *It rules out* new observable coefficients arising from a metric reparametrization alone. *It does not forbid a distinct QGD quantum mechanism. For example, an independently matched covariant correction $\Delta\Gamma_{\rm QGD}[\sigma,\Psi;\hbar]$ could alter the field equations, whether through additional natural-invariant physics, a quantum-derived metric operator, or a consistently specified quantum measure. But a coefficient copied from a known GR effective action, or chosen to reproduce the historical force, would not constitute an independent QGD prediction. A meaningful comparison must specify the same physical source, asymptotic conditions, renormalization prescription, and gauge-invariant observables.*

12.1 An exact exterior test of the historical $9/2$ law

The attractive force candidate in Eq. (48) can be integrated unambiguously at the displayed order. Put $\ell_C = \hbar/(m_f c)$, where the reference species m_f must be specified independently. In the weak-field nonrelativistic interpretation, $F_r = -m_t \partial_r \Phi$ and $\Phi(\infty) = 0$ imply

$$\boxed{\Phi_{9/2}(r) = -\frac{G_N M}{r} - \frac{3 G_N M \ell_C^2}{2 r^3} + \text{higher-order terms.}} \tag{68}$$

For the isolated displayed r^{-3} term and $r > 0$, $\nabla^2(r^{-3}) = 6 r^{-5}$, so

$$\boxed{\nabla^2 \Phi_{9/2}(r) = -\frac{9 G_N M \ell_C^2}{r^5} \neq 0 \quad (r > R),} \tag{69}$$

where R is the support radius of an assumed compact source and higher-order terms have been omitted consistently. If one tried to retain the *unmodified* Poisson equation, the same radial tail would demand an exterior effective density

$$\boxed{\rho_{\text{eff}}(r) = -\frac{9 M \ell_C^2}{4\pi r^5}, \qquad r > R,} \tag{70}$$

which is not the original empty exterior. This is a diagnostic *effective* density, not a claim about negative physical matter. Therefore the historical universal correction cannot be generated solely by changing a compact spherical source profile under the same vacuum Poisson equation; it requires a justified exterior quantum source/response, a modified field equation, or other physics.

At the fully relativistic level, this weak-field calculation is not by itself a complete covariant metric solution. The independently varied Einstein–Hilbert vacuum branch of Chapter 2 already fixes the exact spherical exterior to Schwarzschild. A new universal $1/r^3$ metric perturbation must consequently be checked against the *complete* modified field equations and their conservation and boundary conditions; a scalar force ansatz cannot perform that check. If m_f denotes the test particle mass, its Compton scale also makes the force state/species-dependent, so the claim is not automatically a universal exterior metric correction.

12.2 A falsifiable QGD-specific matching test

A future new term should be expressed as the difference between a microscopically justified QGD effective functional and the specified semiclassical comparator:

$$\boxed{\Delta\Gamma[\sigma, \Psi; \hbar] := \Gamma_{\text{QGD}}[\sigma, \Psi; \hbar] - \Gamma_{\text{sc}}[g[\sigma], \Psi; \hbar].} \tag{71}$$

At fixed physical matter state and boundary data, a nonzero $\Delta\Gamma$ is *necessary but not sufficient* for a new observable: it may contain a boundary term, a redundancy, or an operator removable by permissible redefinitions. The minimum verification gates are:

1. Derive $\Delta\Gamma$ from the stated microscopic/macroscopic Dirac construction or declare any additional physical axiom.

2. Specify its covariant dependence on the full signed master metric, $\sigma^{(s)}$, and any derivative operators; include its independent normalization, units and boundary terms.

3. Vary the *complete* functional, including connection, coframe, stress-energy and possible nonlocal contributions, and check Bianchi/conservation identities and regular branches.

4. Fix at most the permitted Newtonian calibration coefficient, then predict a second observable without re-fitting it.

5. Compare with *both* semiclassical GR plus quantum matter and, where applicable, low-energy quantized-GR effective-field-theory corrections, at matched sources and renormalization convention.

A quantum Dirac radial equation in the unmodified Schwarzschild metric meets none of the first three gates for a new *gravitational field* operator. The previously derived quantum density-gradient stress meets the microscopic matter-side test, but its gravitational consequences within Eq. (7) are also consequences of the chosen semiclassical Einstein–matter system. Hence the present QGD action does *not yet* close the deeper foundational gap of demonstrating that QGD itself predicts a quantum correction independently of semiclassical general relativity.

13 The quantum structure of the master metric (QGD–110)

This section uses only the founding principles: F1, the microscopic Dirac field; F2, coarse-graining and the macroscopic spinor equation; F3, the master metric with its reference term; F4, the adopted Einstein–Hilbert-plus-matter action; F5, the Newtonian calibration. Imported results are marked L.

13.1 A metric-level quantum term and the field equations

Proposition 13.1 (Exact split identity (D))**.** *Write the physical metric as $g = g_{\rm cl} + q$. If $G[g] = 8\pi G\, T$, then identically*

$$G[g_{\rm cl}] = 8\pi G\,(T + T_Q), \qquad T_Q := -\frac{G[g] - G[g_{\rm cl}]}{8\pi G}. \tag{72}$$

A quantum term in the master metric therefore always produces a quantum-corrected Einstein equation for the classical part. By itself this is a rewriting. It acquires content only when F1–F5 fix q, which they do through the quantum stress of the matter.

13.2 The quantum term in the static natural-invariant variables

Proposition 13.2 (Exact identity (D))**.** *In the static branch ($F = 1 - A^2$, $D = 1 + B^2 - A^2$, areal radius),*

$$A^2 - B^2 = 1 - FH, \qquad \frac{\mathrm{d}}{\mathrm{d}r}\ln(FH) = \frac{8\pi r(\rho + p_r)}{1 - 2\mathfrak{m}/r}, \tag{73}$$

with $H = D/F$. Hence $A = B$ exactly whenever $\rho + p_r = 0$: in vacuum, for Maxwell fields and for a cosmological constant.

For the vacuum of the microscopic Dirac field (Proposition 11.1), Eq. (65) gives

$$A^2 - B^2 = -\frac{2}{15\pi}\,\frac{G_N^2 M\hbar}{c^5 r^3}, \tag{74}$$

because vacuum polarization of the microscopic Dirac field has $\rho + p_r < 0$. Inserting this q into Eq. (72) returns $T_Q = -T^{\rm Dirac}$ in all three independent components (verified symbolically). This is the quantum term F1 and F4 place in the master metric. Its scale is ℓ_P^2, not the historical ℓ_Q, and by Theorem 12.1 it coincides with semiclassical GR.

Proposition 13.3 (Test of the historical derivative trial (D))**.** *For the classical exterior sector $A = B = \sqrt{2M/r}$ on the flat spherical reference, the trial $\mathcal{Q}_{ab} = (\bar\nabla_a\sigma^c)(\bar\nabla_b\sigma_c)$ has only angular components, $\mathcal{Q}_{\theta\theta} = 2M/r$, and in areal gauge gives, to first order in M, $\Delta f = 0$ and $\Delta h = -4\lambda M/R^3$. The derived term requires $\Delta f = -4M\ell_P^2/(15\pi R^3)$ and $\Delta h = +2M\ell_P^2/(5\pi R^3)$. No λ matches both, so the trial is not the quantum term of the foundations.*

13.3 A quantum conformal factor derived from the wave equation

Proposition 13.4 (Quantum conformal factor (D)). *For a scalar wave $\psi = Re^{iS/\hbar}$ of mass M obeying $(\Box_g - M^2c^2/\hbar^2)\psi = 0$, the real part of the wave equation is exactly the Hamilton–Jacobi equation of a conformally rescaled metric,*

$$\tilde{g}^{\mu\nu}\partial_\mu S\partial_\nu S + M^2c^2 = 0, \qquad \tilde{g}_{\mu\nu} = \Omega_\psi^2 g_{\mu\nu}, \qquad \Omega_\psi^2 = 1 - \lambda_C^2\frac{\Box_g R}{R}, \quad \lambda_C = \frac{\hbar}{Mc}. \tag{75}$$

Wherever $\Omega_\psi^2 > 0$, the integral curves of $\tilde{g}^{\mu\nu}\partial_\nu S$ are timelike geodesics of $\tilde{g}$.

Proof. The real part is $g^{\mu\nu}\partial S\partial S + M^2c^2 = \hbar^2\Box_g R/R$ (verified symbolically). For the second statement, $u_\mu = \partial_\mu S/(Mc)$ is a gradient with $\tilde{g}(u,u) = -1$, so $u^\nu\tilde{\nabla}_\nu u_\mu = \frac{1}{2}\partial_\mu\tilde{g}(u,u) = 0$. □

This is the quantum potential of Chapter 1 (Eq. wkb-quantumHJ) and of Section 9, in geometric form. Calling the integral curves *trajectories* is an interpretation. The theorem itself is a statement about the wave's phase and amplitude.

13.4 How the quantum term enters the master metric

Theorem 13.5 (Two consistent routes, one inconsistent (D|A)). *Under F1–F5:*

1. ***Per wave (consistent).*** *Ω_ψ^2 multiplies the universal master metric for that wave (spin-0 matter; spin-$\frac{1}{2}$ matter in Section 13.5), $\tilde{g}^{(\psi)}_{\mu\nu} = \Omega_\psi^2 T^a{}_\mu T^b{}_\nu[\eta_{ab} + \sum_s s_s\sigma_a^{(s)}\sigma_b^{(s)}]$, and governs that wave's phase dynamics. Universal free fall is recovered as $\lambda_C \to 0$; at finite λ_C the mass dependence is that of ordinary quantum mechanics in a gravitational field.*

2. ***Through the source's stress tensor (consistent, universal).*** *On shell the wave's stress tensor splits exactly as*

$$T_{\mu\nu} = \rho\, u_\mu u_\nu + T^Q_{\mu\nu}, \qquad T^Q_{\mu\nu} = \hbar^2\big[\partial_\mu R\partial_\nu R - \tfrac{1}{2}g_{\mu\nu}\big((\partial R)^2 + R\Box_g R\big)\big], \tag{76}$$

   with $\rho = M^2c^2R^2$ and the on-shell $u_\mu = \partial_\mu S/(Mc)$. Under F4, T^Q curves spacetime with coupling $8\pi G$, and through Eq. (72) it appears as a universal quantum sector q of the master metric. For a coherent static source of size L, $T^Q_{00}/\rho c^2 = O(\lambda_C^2/L^2)$.

3. ***Source's conformal factor in the universal metric (inconsistent).*** *For $g = (1+\varepsilon)g_{\rm cl}$, the linearized Einstein tensor $G_{00} = -\nabla^2\varepsilon$ carries no factor of G; for a Gaussian amplitude it equals $6\lambda_C^2/L^4$. This contradicts F4 at order $\hbar^2G^0$, and would require replacing F4.*

The result is a two-level master metric derived from the founding principles:

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu\Big[\eta_{ab} + \sum_s s_s\sigma_a^{(s)}\sigma_b^{(s)} + q_{ab}\Big], \qquad \tilde{g}^{(\psi)}_{\mu\nu} = \Omega_\psi^2\, g_{\mu\nu},} \tag{77}$$

with sector signs $s_s = -\varepsilon_s$ in the notation of Eq. (6), and where q is fixed by F4 from the quantum stress of the source (vacuum polarization, Eq. (74), and the coherent-wave stress T^Q), and Ω_ψ by Eq. (75). This gives the founding master metric's quantum slot, historically $-\lambda\mathcal{Q}$, a derived occupant. As $\hbar \to 0$, q's quantum part and $\Omega_\psi - 1$ vanish, recovering the classical restriction and every classical result of Chapters 2–4. The structure reproduces quantum mechanics in curved spacetime and semiclassical gravity, and it does not quantize the metric: the limitations of semiclassical gravity, for example for macroscopic superpositions of a source, apply unchanged.

Proposition 13.6 (No natural-invariant dynamics beyond g under F1–F5 (D)). *The microscopic Dirac action uses a vierbein of g and is locally Lorentz invariant (Chapter 2, Clifford compatibility), so it depends only on g. The Maxwell, scalar-contact and Einstein–Hilbert terms and the quantum matter effective action $\Gamma[g]$ are functionals of g. Theorem 12.1 therefore applies: dynamics acting on more than g requires an additional founding principle.*

13.5 The per-wave law for spin-$\frac{1}{2}$ matter (QGD–112, QGD–113)

For spin-$\frac{1}{2}$ matter no metric conformal to g makes the flow lines geodesic (QGD–111). The exact replacement follows from the macroscopic spinor equation alone. Define $\sigma = \bar{\psi}\psi \neq 0$, the convective momentum $P_\mu = -\operatorname{Im}(\bar{\psi}\nabla_\mu\psi)/\sigma$, the spin remainder $\Lambda = \nabla\bar{\psi}\cdot\nabla\psi - |\bar{\psi}\nabla\psi|^2/\sigma$, the spin tensor $S^{ab} = \bar{\psi}\Sigma^{ab}\psi$ with $\Sigma^{ab} = \frac{i}{2}[\gamma^a, \gamma^b]$, and the spin vorticity $F = \mathrm{d}P$.

Theorem 13.7 (Spin-$\frac{1}{2}$ flow law in a curved universal metric (D))**.** *In Chapter 1's conventions,* $(\gamma^\mu\nabla_\mu)^2 = -\Box + \frac{1}{4}R$ *and* $[\nabla_\mu, \nabla_\nu] = -\frac{1}{4}R_{ab\mu\nu}\gamma^a\gamma^b$ *on spinors. Consequently, exactly,*

$$P\cdot P + M^2 + \tfrac{1}{4}R = \frac{\Box\sqrt{\sigma}}{\sqrt{\sigma}} - \frac{\Lambda}{\sigma}, \qquad F_{\mu\nu} = -\frac{\operatorname{Im}(\nabla_{[\mu}\bar{\psi}\nabla_{\nu]}\psi)}{\sigma} - \frac{\partial_{[\mu}\sigma\, P_{\nu]}}{\sigma} - \frac{R_{ab\mu\nu}S^{ab}}{4\sigma}, \tag{78}$$

where $X_{[\mu\nu]} := X_{\mu\nu} - X_{\nu\mu}$*; and with* $\tilde{g} = \Omega_{1/2}^2 g$, $\Omega_{1/2}^2 = 1 + \frac{R}{4M^2} - \frac{1}{M^2}\left(\frac{\Box\sqrt{\sigma}}{\sqrt{\sigma}} - \frac{\Lambda}{\sigma}\right)$ *and* $\tilde{u}_\mu = P_\mu/M$,

$$\boxed{M\,\tilde{u}^\nu\tilde{\nabla}_\nu\tilde{u}_\mu = \tilde{u}^\nu F_{\nu\mu}.} \tag{79}$$

The convective flow moves in the conformally rescaled metric like a charge in the spin vorticity. F contains a curvature–spin term with the structure of the Mathisson–Papapetrou coupling, derived here rather than imported. For spin-0 matter $\Lambda = 0$ and $F = 0$, and the law reduces to Proposition 13.4.

Verification. Flat-region identities verified to 10^{-14} on exact superpositions of plane-wave solutions (QGD–112). The curvature coefficients $+\frac{1}{4}$ and $-\frac{1}{4}$, the curved off-shell identity and the curved spin vorticity were verified on an FLRW metric with an arbitrary spinor field, residuals $\leq 6\times10^{-16}$, with the spinor connection constructed from $\nabla\gamma = 0$ (QGD–113). In generic states the spin-vorticity force is 40–70% of the gradient force.

Slow-motion limit (QGD–113). With $\hat{s}$ the spin direction, $\Lambda/\sigma \to \frac{1}{4}\partial\hat{s}\cdot\partial\hat{s}$ (from the exact Fubini–Study identity for unit two-spinors) and $F_{ij} \to -\frac{1}{2}\hat{s}\cdot(\partial_i\hat{s}\times\partial_j\hat{s})$, a Mermin–Ho-type relation, both with $O(v^2)$ convergence verified. The kinetic energy per particle is $p^2/2M - (\hbar^2/2M)\nabla^2\sqrt{n}/\sqrt{n} + (\hbar^2/8M)|\nabla\hat{s}|^2$. For constant spin the law reduces exactly to Chapter 1's irrotational result $\dot{\boldsymbol{p}} = -\nabla(U+Q)$ and to the fixed-spin quantum stress of Section 9. When spin varies, a spin quantum potential and a spin-vorticity force appear that those fixed-spin treatments omit.

Closed form of the spin-$\frac{1}{2}$ stress (QGD–123). Using the macroscopic spinor equation on ψ and on $\bar{\psi}$, with $\gamma^\lambda\gamma^\mu = -\eta^{\lambda\mu} - i\Sigma^{\lambda\mu}$, the symmetric (Belinfante) stress tensor in a flat region is exactly

$$T^{\mu\nu} = \frac{\sigma}{M}P^\mu P^\nu + \frac{1}{2M}\left(\partial^\mu\sqrt{\sigma}\,\partial^\nu\sqrt{\sigma} - \sqrt{\sigma}\,\partial^\mu\partial^\nu\sqrt{\sigma}\right) + \frac{1}{M}\Lambda^{\mu\nu} - \frac{1}{4M}\partial_\lambda\operatorname{Im}\left(\bar{\psi}\Sigma^{\lambda\mu}\partial^\nu\psi + \bar{\psi}\Sigma^{\lambda\nu}\partial^\mu\psi\right), \tag{80}$$

with the tensor spin remainder $\Lambda^{\mu\nu} = \operatorname{Re}(\partial^\mu\bar{\psi}\partial^\nu\psi) - \operatorname{Re}(\bar{C}^\mu C^\nu)/\sigma$, whose trace is Λ. The terms are: dust; a scalar quantum stress; a spin remainder; and a spin curl, which is separately conserved because $\Sigma^{\lambda\mu}$ is antisymmetric. Verified on exact superpositions of plane-wave solutions (relative residual $\leq 5\times10^{-13}$). Relative to dust, the scalar and spin-remainder terms are second order in λ_C/L (slopes 1.92, 1.94) and the spin curl is first order (slope 1.02). The spin curl is the exact origin of the first-order tilt between probability flow and energy flow. With $\hbar$ restored, the quantum stress of the universal level for spin-$\frac{1}{2}$ matter is therefore given in closed form. The curved-metric version (replacing ∂ by the spinor covariant derivative) adds curvature terms that have not been derived here.

Conditions. $\sigma \neq 0$ and P timelike. As for spin-0 matter, the law describes the convective momentum field; reading its integral curves as trajectories is an interpretation.

13.6 Where the natural invariant comes from

Proposition 13.8 (Closure identity (D))**.** *On the static one-null-sector branch* $g = \bar{g} + H\,\mathrm{d}v^2$ *with* $H \to 0$ *at infinity, wherever* $0 < H < 1$*,*

$$\sigma_{\mathrm{amp}} = \sqrt{H} = \beta_{\mathrm{ff}} = \frac{2\zeta_{\mathrm{ff}}}{1+\zeta_{\mathrm{ff}}^2}, \qquad \zeta_{\mathrm{ff}} = \frac{\sqrt{H}}{1+\sqrt{1-H}}, \tag{81}$$

where β_{ff} *is the speed, relative to static observers, of a test particle released from rest at infinity, and* ζ_{ff} *is its Chapter 1 on-shell spinor ratio. Covariantly,* $\sigma_{\mathrm{amp}}^2 = 1 - (u_{\mathrm{ff}} \cdot u_{\mathrm{stat}})^{-2}$*. The sector direction* $k = \mathrm{d}(t+r)$ *is a phase gradient solving the massless limit of Chapter 1's Hamilton–Jacobi equation.*

This extends Chapter 2's Schwarzschild diagnostic to the whole static one-null-sector class. It fails for Kerr: the free-fall speed differs from the Kerr–Schild amplitude, and the oblate k is not closed. It also fails where $FH \neq 1$, for example in Eq. (74). It establishes self-consistency, not origin. Chapter 1 (Proposition QGD–116) uses it to repair the historical source profile: the historical $b_f = \sqrt{f}$ is the velocity, and the spinor ratio is $\zeta_f = \sqrt{f}/(1+\sqrt{1-f})$, which matches exact radial solutions to 3×10^{-4}. The natural invariant's algebraic *form* is derived from the macroscopic spinor equation, while its exterior *values* for a given source are selected by F4.

13.7 Lorentz covariance and gauge dependence

The main equations were tested directly (QGD–110):

Equation or object	Result
Master metric, Eq. (6)	Invariant under frame Lorentz transformations ($T \mapsto \Lambda T$, $\sigma_a \mapsto \sigma_c(\Lambda^{-1})^c{}_a$); tensorial under coordinate changes
Historical $\mathcal{Q}$ with $\bar{\nabla}$	Tensorial: Cartesian and spherical computations agree exactly
Macroscopic spinor equation (Ch. 1 conventions)	Covariant: a spinor representation $S(\Lambda)$ exists and $Su(p)$ solves the equation at Λp
Spinor ratio $\mathsf{R}(p)$	Transforms exactly as $\mathsf{R}(p) \mapsto \mathsf{R}(\Lambda p)$
$\zeta = p/(E + Mc^2)$	Poincaré-ball coordinate of the mass shell; boosts act by Möbius addition (exact). $\beta = 2\zeta/(1+\zeta^2)$ is the Beltrami–Klein coordinate
Spinor component moduli, $\psi^\dagger\psi$	**Not** invariant; $\bar{\psi}\psi$ and $j^\mu j_\mu$ are
Ω_ψ, $T^{\mathcal{Q}}$ (scalar wave)	Covariant
Sector representation σ	**Gauge dependent**: the same Schwarzschild geometry (Kretschmann $12r_s^2/r^6$ in both) has $h = g - \bar{g}$ of rank 1 in the advanced identification and rank 2 in the static one

Three consequences follow:

1. **Spin-$\frac{1}{2}$ version of the quantum terms (QGD–111–113).** The scalar-wave Eqs. (75) and (76) use component amplitudes, which are not invariant for spinors. The covariant spin-$\frac{1}{2}$ amplitude is $(-j \cdot j)^{1/4}$ (Fierz identity, verified). The per-wave level for spin-$\frac{1}{2}$ matter is Theorem 13.7. The universal level holds unchanged: in the energy frame, the stress tensor of a coherent spin-$\frac{1}{2}$ field is $\rho\, u_L u_L + \Pi$ with $\Pi = O((\lambda_C/L)^2)\rho$ (numerical scaling, slope 1.98), and the probability current tilts relative to the energy flow at $O(\lambda_C/L)$ (slope 1.01) through the spin current, which is not an extra stress.

2. **Geometric meaning of the spinor ratio.** The Chapter 1 ratio ζ is not a Lorentz vector. It is a hyperbolic-geometry coordinate (Poincaré ball), and the closure identity (81) is the ball-to-Klein map.

3. **The natural invariant is not a diffeomorphism invariant.** Because the master metric contains a fixed reference $\bar{g}$, the sector representation, including the number of sectors, their rank and their signs, depends on how spacetime is identified with the reference. Only g, and quantities built from it, are gauge invariant. Every QGD prediction under F4 is computed from g, so no prediction is affected, but statements about σ must state the identification used.

14 What the equation makes possible—and what it does not

Equation (12) has an important explanatory role. It expresses effective gravitational equilibrium as a balance of the metric-mediated gravitational action against physical matter stress, projected into the natural-invariant variables. The source is the *complete, conserved* stress of the physical system; a local Dirac-component ratio or a single wave amplitude is not generally an adequate universal source. The Dirac state may contribute through basis-independent currents, energies and stress correlations.

But the action-derived equation also establishes a *limit*: with no direct natural-invariant interactions and no new dynamical operator, the effective field equations contain no distinct propagating σ mode beyond the effective metric degrees of freedom. A new observable could follow from (i) a well-defined direct coupling between quantum matter and natural-invariant data, (ii) a physically derived independent natural-invariant kinetic/constraint sector, or (iii) a genuinely distinct quantum effective action. None follows inevitably from the signed metric identity; each would require a separately specified covariant action, conservation laws, correct field content and measurable consequences.

Note on route (i) (QGD–100, corrected QGD–105). Research note QGD–100 examined one *hypothetical* realization of route (i): removing the reference term, reading four η-signed sectors as a tetrad, and assuming a microscopic theory with only global frame symmetry. Neither assumption belongs to the formulation of Chapters 1–4. Under them the two-derivative dynamics is New General Relativity [11], whose parameter directions other than the axial one give $\gamma = -2c/(a+b-4c) \neq 1$ at linear order, and whose axial direction is obstructed at nonlinear order according to [17]. This rules out that particular hypothetical route; it says nothing about routes (i)–(iii) in general.

A particularly sharp discriminator is the comparison of different physical quantum source states with matched mean stress and classical boundary data. A measured mean-field difference beyond standard GR plus quantum-field corrections would indicate a new physical coupling. Merely assigning different signed representatives to the same metric cannot change a metric-only observable.

15 Relationship to the archived historical Chapter 4

The historical research program wrote schematically

$$\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu + \cdots . \tag{82}$$

This is *not* equivalent by definition to Eq. (12). The historical Q_μ, G_μ, T_μ expressions have unresolved algebraic/provenance and normalization issues documented in the QGD–052 audit. In particular, no term $\Box_g \sigma_\mu$ or $\Box_g^2 \sigma_\mu$ can be deduced simply from varying the derivative-free composite Einstein–Hilbert action. Adding a standalone σ kinetic term would be a change of the effective theory and requires physical justification, not a cosmetic rewriting of the balance equation. The archived manuscript therefore remains an independent historical source, while the present chapter begins with the precise equation supported by the accepted action and metric map.

16 Conclusion and the next legitimate question

The action-based content of this updated Chapter 4 has one central field-equation result:

$$\boxed{\mathscr{S}^{\rho}_{(s)} = 2s_s\mathcal{E}_g^{\rho\nu}\sigma_\nu^{(s)} = 0, \qquad \mathscr{S}^{\rho}_{(s)} = \mathscr{G}^{\rho}_{(s)} + \mathscr{T}^{\rho}_{(s)} + \mathscr{Q}^{\rho}_{(s)}.} \tag{83}$$

For the metric-only effective action, $\mathscr{Q} = 0$ and, on the full-rank signed branch, this balance reproduces the established Einsteinian field equations. This is a rigorous mathematical bridge from QGD's natural-invariant variables to effective gravity, not a claim that the underlying physical origin of gravity has been solved. The question that can make QGD more than a geometric reformulation is therefore precise: *Does the quantum matter-wave structure lead to independent, covariantly defined natural-invariant dynamics or source dependence, and does any such effect produce an observable not already predicted by GR and quantum field theory?*

The Newton-calibrated Chapter 3 potential and the controlled $\hbar^2$ quantum-source stress are compatible with this balance, but neither supplies an independent natural-invariant propagation operator. The metric-only quantum-equivalence theorem in Section 12 makes this limitation exact under its stated rank and action assumptions; the historical attractive 9/2 force would additionally require a nonzero exterior source residual or a changed gravitational response. The historical 9/2 correction remains unverified. Future extensions must derive their covariant operators, coefficient matching and physical observables before assigning a genuinely new wave equation or degree of freedom to σ_μ.

Revision history

QGD–096 established the equivalence/no-independent-correction theorem for the regular metric-only theory and calculated the nonzero vacuum-Poisson residual that the historical attractive 9/2 force would require. **QGD–097** added a separately sourced semiclassical benchmark for a massless four-component Dirac vacuum field: a verified far-field $-2G_N^2M\hbar/(15\pi c^3r^3)$ correction to the static Killing-energy potential, taken from published quantum-matter backreaction, not from the historical complex-phase Taylor expansion or a new QGD operator. **QGD–098** independently derived and checked the angular-channel coefficients of the Schwarzschild radial Dirac reduction and recorded why its Liouville-normal effective potential cannot be promoted to a physical force by differentiation. **QGD–101** integrated research notes QGD–099 and QGD–100. **QGD–105** audited those insertions: Remark 5.2 was restricted to Chapter 2's constructed branch; the route-(i) paragraph was restated as a test of one hypothetical realization; and a closing paragraph that reframed QGD's identity was removed, because it did not follow from the chapter's stated assumptions. No earlier equation or label was removed. **QGD–110** added Section 13: the split identity, the quantum term in static variables, the test of the historical derivative trial, the derived quantum conformal factor, the two-level master metric, the closure identity, and Lorentz-covariance and gauge tests. **QGD–111** replaced the open spin-$\frac{1}{2}$ item with covariant results; **QGD–112** added the exact spin-$\frac{1}{2}$ flow law and restricted the name "Dirac" to the microscopic field. **QGD–113** added the curved-metric form of that law (Theorem 13.7), its slow-motion limits, and the cross-reference in Section 9. **QGD–123** added the closed form of the spin-$\frac{1}{2}$ stress, Eq. (80), and applied the naming convention to derived objects. **QGD–124** added the abstract sentence on QGD's place as quantum gravity. **QGD–120** added Section 3.2: the complete QGD action, its symmetries and its quantum effective form. **QGD–121** added Section 4.5: the degree-of-freedom count (two propagating modes on spanning branches; degenerate and non-spanning cases). **QGD–122** added Proposition 4.6: spanning representations of the chapters' black-hole and cosmological backgrounds, and the projection form of one-sector reductions. **QGD–115** added Section 3.1: the Einstein–Hilbert form derived by uniqueness under Principle L, with the coupling from the Newtonian calibration.

Source and research provenance

References

[1] QGD, *Chapter 1: Foundations*, accepted Dirac/matter-wave construction (existing research manuscript).

[2] QGD, *Chapter 2: Field Equations*, complete signed master metric, variational right-inverse and effective action (QGD–075).

[3] QGD, *Chapter 3: Gravitational Normalization and Quantum-Force Verification* (QGD–087 surgical revision; QGD–101 notation update).

[4] QGD–052, *Focused Historical QGT/QGD Field-Equation Provenance Audit*; and QGD–094, *General Natural-Invariant Field Equations and Historical Source Split*, working research checks.

[5] QGD–091–092, *Independent Schwarzschild Dirac radial reduction and referee verification*, conditional test-field coefficient audit (cross-checked in QGD–098).

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17 The quantum correction slot: what QGD actually derives

Scope and proof status. The complete founding master metric is

$$\boxed{g_{\mu\nu} = T^{a}{}_{\mu} T^{b}{}_{\nu} \left[\eta_{ab} - \sum_{s} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right].} \tag{84}$$

Here the historical $\mathcal{Q}_{ab}$ is a proposed independent metric-derivative operator. We distinguish it from (i) a computed, state-dependent quantum metric deformation $q_{ab}^{\rm back}$ and (ii) the phase/amplitude quantum potential of an individual matter wave. We prove a linear response identity and a metric-level non-identifiability statement; neither constitutes a quantum microscopic derivation of a unique new $\mathcal{Q}[\sigma]$ operator.

17.1 The response that follows from the renormalized action

Let $g^{(0)}$ solve the selected classical effective Einstein–matter problem and assume a controlled perturbative state with $\langle \widehat{T}_{\mu\nu} \rangle_{\rm ren} = O(\hbar)$. For $g = g^{(0)} + q^{\rm back} + O(\hbar^2)$, expansion of the quantum-corrected metric equations at first order yields

$$\boxed{\mathcal{L}_{g^{(0)}}[q^{\rm back}]_{\mu\nu} = \frac{8\pi G_N}{c^4} \langle \widehat{T}_{\mu\nu} \rangle_{\rm ren}^{(1)}.} \tag{85}$$

The linear operator $\mathcal{L}_{g^{(0)}}$ includes the linearized classical geometric and matter Euler derivatives; if quantum polarization is resummed, the corresponding causal stress response kernel must also be included. In a chosen gauge and with admissible boundary/initial data, the formal first-order solution is

$$\boxed{q^{\rm back} = \frac{8\pi G_N}{c^4} \mathcal{L}^{-1}_{g^{(0)},\,\rm bc} \langle \widehat{T} \rangle_{\rm ren}^{(1)}.} \tag{86}$$

For real-time evolution the inverse is a retarded, gauge-constrained response, not an arbitrary Feynman Green function. This is a *derived quantum metric response conditional on the Chapter 4 action*, not a new off-shell operator in the defining metric.

Proposition 17.1 (On-shell historical-slot assignment, convention-dependent)**.** *Fix a reference coframe, a gauge/coordinate identification, a classical natural-invariant representative, a quantum state and boundary data. Choose $\lambda \neq 0$. If the sector configuration is held fixed while representing the first-order quantum correction, one may define*

$$\boxed{\mathcal{Q}_{ab}^{\rm rep} := -\frac{1}{\lambda} (T^{-1})^{\mu}{}_{a} (T^{-1})^{\nu}{}_{b}\, q_{\mu\nu}^{\rm back}.} \tag{87}$$

Then $-\lambda T^{a}{}_{\mu} T^{b}{}_{\nu} \mathcal{Q}_{ab}^{\rm rep} = q_{\mu\nu}^{\rm back}$ by construction. This is an on-shell representation *of a computed metric correction; it is not a unique or autonomous functional $\mathcal{Q}_{ab}[\sigma, \bar{\nabla}\sigma]$ and does not supply the adjoint $D_s\mathcal{Q}$ for a new field equation.*

17.2 A calculated far-field example: one massless Dirac field

Use the Boulware-state far-field solution and the static, fixed-areal-radius coordinates of Chapter 4. Define the length scales $\mu := G_N M/c^2$ and $\ell_P^2 := G_N \hbar/c^3$, and $f_0 := 1 - 2\mu/r$. The one-loop massless-Dirac solution gives, at leading quantum order for $r \gg 2\mu$,

$$f(r) = f_0 - \frac{4\mu\ell_P^2}{15\pi r^3} + \cdots, \qquad f_r(r) := 1 - \frac{2G_N \mathfrak{m}(r)}{c^2 r} = f_0 - \frac{2\mu\ell_P^2}{5\pi r^3} + \cdots. \tag{88}$$

The common-coordinate metric difference relative to classical Schwarzschild is therefore

$$\boxed{q_{00}^{\rm back} = \frac{4\mu\ell_P^2}{15\pi r^3}, \qquad q_{rr}^{\rm back} = \frac{2\mu\ell_P^2}{5\pi r^3} + O\!\left(\frac{\mu^2\ell_P^2}{r^4}\right), \qquad q_{\theta\theta}^{\rm back} = q_{\phi\phi}^{\rm back} = 0.} \tag{89}$$

Here $x^0 = ct$ and the covariant metric components are used; the angular components vanish in areal gauge. Other choices of gauge change components, not physical curvature observables. Selecting, only as a bookkeeping convention, $\lambda = \ell_P^2$, and the diagonal flat spherical reference coframe, Eq. (87) yields the explicit representative

$$\boxed{\mathcal{Q}_{00}^{\rm rep} = -\frac{4\mu}{15\pi r^3}, \qquad \mathcal{Q}_{11}^{\rm rep} = -\frac{2\mu}{5\pi r^3} + O\!\left(\frac{\mu^2}{r^4}\right), \qquad \mathcal{Q}_{22}^{\rm rep} = \mathcal{Q}_{33}^{\rm rep} = 0.} \tag{90}$$

The representative has the required dimension L^{-2} if the natural invariants are dimensionless and λ has dimension L^2. It vanishes from the metric as $\hbar \to 0$, provided the state and classical background have a regular classical limit. No coefficient of an independent new QGD force has been obtained.

Proposition 17.2 (No unique $\mathcal{Q}$ from metric-only dynamics)**.** *At any point with a nondegenerate reference coframe, the complete real signed-sector construction can represent an arbitrary symmetric metric deformation. Therefore a specified total metric and a metric-only effective action do not select a unique additional tensor $\mathcal{Q}_{ab}$. Even with the quantum metric $q^{\rm back}$ computed, one may use an on-shell representative with the historical slot $-\lambda\mathcal{Q} = q^{\rm back}$ and fixed classical sectors,* or *readjust sufficiently many signed sectors and take $\mathcal{Q} = 0$, obtaining the same metric.*

Proof. Write the total internal deformation as $h_{ab} = (T^{-1}gT^{-\mathsf{T}})_{ab} - \eta_{ab}$. For any symmetric h, choose $b \neq 0$ and the signed matrices $P = bI + h/(4b)$ and $N = bI - h/(4b)$. They obey $P^{\mathsf{T}}P - N^{\mathsf{T}}N = h$ exactly. Their rows supply a real signed-sector representation with $\mathcal{Q} = 0$. Alternatively, retain a chosen classical set of sectors and assign $-\lambda\mathcal{Q}$ to the remaining deformation. Because the metric-only action sees only the resulting g, it does not discriminate these metric representations. The argument is algebraic and on-shell: it does not establish equivalence of all *off-shell* derivative-dependent changes of fields or their quantum functional measures. □

17.3 Why a historical local derivative candidate is not enough

The trial $\mathcal{Q}_{ab}^{\rm trial} = (\bar\nabla_a\sigma^c)(\bar\nabla_b\sigma_c)$ has the correct schematic dimensional order: on the Schwarzschild background $\sigma \sim \sqrt{\mu/r}$ and $\ell_P^2\mathcal{Q}^{\rm trial} \sim \mu\ell_P^2/r^3$. But Chapter 4's independent tensor test found that no one coefficient reproduces *both* the quantum temporal and radial metric corrections. Matching a radial power law does not match a covariant tensor. The known renormalized matter effective action also contains state-dependent and generally nonlocal responses, which a single local first-gradient ansatz need not reproduce.

Consequence for Option C. For a genuinely specified differentiable off-shell tensor $\mathcal{Q}[\sigma; T]$, the preferred QGD equation is

$$\boxed{2s_s\Pi^{ab}\sigma_b^{(s)} = \lambda\big[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\big]^a, \qquad s_s = -\varepsilon_s.} \tag{91}$$

But substituting the on-shell representative (87) into this equation as if it were an independent operator would generally double count or misrepresent the same quantum backreaction already present in $\hbar\Gamma^{(1),\rm ren}_{\rm matter}$. The presently controlled dynamical formulation sets the independent historical slot to zero and retains its quantum source in Π^{ab}. A nonzero new $\mathcal{Q}$ requires an independently defined sector-sensitive observable, a microscopic matching calculation, or a nonredundant dynamical principle.

18 Energy charges and the meaning of the projected field equations (QGD–139)

Premises and scope. This section uses the complete founding metric $g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}]$. The explicit calculations are on its established fixed-coframe, $\lambda\mathcal{Q} = 0$ metric-only branch. The historical $\mathcal{Q}$ remains uncalculated; a background translation current is not thereby proved to be a unique gauge-invariant local gravitational stress tensor.

18.1 Two levels of energy conservation

For Chapter 1's flat-space matter action, translation invariance implies $\partial_\mu \Theta^\mu{}_\nu = 0$ on the microscopic matter equations. For Chapter 4's emergent effective metric, the covariant Bianchi identity and metric matter equations give $\nabla_\mu T^{\mu\nu}_{\text{eff}} = 0$, including the renormalized quantum source in its regime of validity. They are not the same local current. Where ξ^ν is a Killing field, $J^\mu := T^{\mu\nu}_{\text{eff}} \xi_\nu$ obeys

$$\nabla_\mu J^\mu = 0, \qquad \partial_\mu(\sqrt{-g}\, J^\mu) = 0. \tag{92}$$

This is a matter Killing-energy balance; in a gravitating system the Hamiltonian energy additionally includes a gravitational boundary term. For nonstationary systems an asymptotic, e.g. Bondi, flux replaces a global stationary Killing energy.

18.2 The Option C equation and the conservation constraints

For the metric-only functional, define $\Pi^{ab} := T^a{}_\mu T^b{}_\nu \mathcal{E}_g^{\mu\nu}$, where $\mathcal{E}_g^{\mu\nu} = (\sqrt{-g})^{-1}\delta\Gamma/\delta g_{\mu\nu}$. The complete metric-map projection is

$$\boxed{2 s_s \Pi^{ab} \sigma_b^{(s)} = \lambda[(D_s \mathcal{Q})^*_{\sqrt{-g}} \Pi]^a, \qquad s_s = -\varepsilon_s.} \tag{93}$$

On a regular branch with a surjective complete metric Jacobian, Eq. (93) is equivalent to $\Pi = 0$, that is the quantum-corrected Einstein–matter equations. Thus both displayed sides vanish on shell: they are *not* separately observable opposing energy-transfer channels. The energy conservation law is instead obtained from the diffeomorphism/translation Noether identities and boundary Hamiltonian structure. A new physical sector energy requires a separately justified sector-sensitive action, which the established theory lacks.

18.3 Schwarzschild: matching the natural-invariant flux to ADM mass

Let $H = A^2 = B^2 = 2G_N M/(c^2 r) = C/r$ for the standard flat-reference Schwarzschild Kerr–Schild exterior. On a t = const Kerr–Schild slice in asymptotically Cartesian coordinates, $\gamma_{ij} = \delta_{ij} + H n_i n_j$, $n_i = x_i/r$. Since

$$\partial_j(\gamma_{ij} - \delta_{ij}) - \partial_i(\gamma_{jj} - 3) = \frac{2H}{r}\, n_i \qquad \text{for } H = C/r, \tag{94}$$

the ADM mass is

$$\boxed{M_{\text{ADM}} = \frac{c^2}{16\pi G_N} \lim_{r\to\infty} \oint \left[\partial_j \gamma_{ij} - \partial_i \gamma_{jj}\right] n^i \, dS = \frac{c^2 C}{2G_N} = M.} \tag{95}$$

Independently, Chapter 5's accepted static energy gives $\nabla^2 H = -8\pi G_N \rho_m/c^2$ and hence its source flux

$$\boxed{M_{\text{flux}} := -\frac{c^2}{8\pi G_N} \oint \partial_r H \, dS = \frac{c^2 C}{2G_N} = M_{\text{ADM}}.} \tag{96}$$

This is an exact *exterior-charge correspondence* for the named Kerr–Schild branch and asymptotic conditions; identifying the Newtonian source integral with a fully relativistic interior ADM mass requires the appropriate internal, kinetic and binding contributions, not simply $\int \rho_m \, d^3x$ beyond the Newtonian order.

18.4 Exact spherical mass as a function of the natural invariants

For Chapter 2's one-sector spherically symmetric metric in an areal radius r, put $F = 1 - A^2$ and $D = 1 + B^2 - A^2$. The inverse radial metric coefficient is $g^{rr} = F/D$, independent of the static off-diagonal $dt\,dr$ representation. The standard Misner–Sharp mass therefore has the exact QGD representation

$$\boxed{m_{\rm MS}(r) = \frac{c^2 r}{2G_N}\big(1 - g^{ab}\partial_a r \partial_b r\big) = \frac{c^2 r}{2G_N}\frac{B(r)^2}{1 + B(r)^2 - A(r)^2}.} \tag{97}$$

This is an identity in the specified spherically symmetric metric representative, not a new GR mass or a unique local stress tensor. On $A^2 = B^2 = H = 2G_N M/(c^2 r)$ it gives $m_{\rm MS} = M$ identically in the vacuum exterior. In a static Einstein–matter solution its radial derivative satisfies the geometric-source balance $m'_{\rm MS}(r) = 4\pi r^2 \epsilon(r)/c^2$ for total local rest-frame energy density ϵ in the conventional static spherical frame, including renormalized quantum contributions when these are consistently included.

In Chapter 4's specific massless-Dirac asymptotic calculation, $A^2 = H + \delta A^2$, $B^2 = H + \delta B^2$, where $\delta A^2 = 4G_N^2 M\hbar/(15\pi c^5 r^3)$ and $\delta B^2 = 2G_N^2 M\hbar/(5\pi c^5 r^3)$. First-order expansion gives

$$\boxed{\delta m_{\rm MS} = \frac{c^2 r}{2G_N}\left[(1-H)\delta B^2 + H\delta A^2\right] = \frac{G_N M\hbar}{5\pi c^3 r^2} + O\!\left(\frac{G_N^2 M^2 \hbar}{c^5 r^3}\right).} \tag{98}$$

It reproduces the independently calculated radial mass function of Chapter 4 (geometrized $m(r) = M[1 + 1/(5\pi r^2)] + \cdots$). The exterior quantum stress can make this finite-radius mass larger than its fixed asymptotic ADM value; $m'_{\rm MS}$ is then negative in that state. This is a check of the effective equations, not a new source-dependent quantum force from sector reparameterization.

18.5 An exact static-spherical QGD to Brown–York energy dictionary

For Chapter 2's general one-sector static spherical metric, write $\sigma = A(r)\,d(ct) + B(r)\,dr$ and define $F := 1 - A^2$, $D := 1 + B^2 - A^2$. After removal of the static cross term, the radial component on the orthogonal static slice is $\gamma_{rr} = D/F$. The flat-referenced Brown–York energy of a round sphere with areal radius R therefore becomes

$$\boxed{E_{\rm BY}(R) = \frac{c^4 R}{G_N}\left[1 - \sqrt{\frac{1 - A(R)^2}{1 + B(R)^2 - A(R)^2}}\right], \quad F, D > 0.} \tag{99}$$

Proof: On this spatial slice, the outward mean curvature is $k = (2/R)\sqrt{F/D}$; the flat reference has $k_0 = 2/R$. The Brown–York Hamiltonian surface integral $E = (c^4/8\pi G_N)\oint (k_0 - k)\,dS$ reduces to Eq. (99). For the exterior classical branch $A^2 = B^2 = H$ this reduces to Eq. (101). It is a quasilocal mass in the specified static slice and reference; for other slicings and observers the Brown–York energy differs.

For the massless-Dirac far-field backreaction, let $A^2 = H + \delta A^2$ and $B^2 = H + \delta B^2$ with $H = 2G_N M/(c^2 R)$ and $\delta A^2 = 4G_N^2 M\hbar/(15\pi c^5 R^3)$, $\delta B^2 = 2G_N^2 M\hbar/(5\pi c^5 R^3)$. Differentiating Eq. (99) to first order in $\delta A^2, \delta B^2$ gives

$$\boxed{\delta E_{\rm BY}(R) = \frac{c^4 R}{2G_N}\sqrt{1-H}\left[\delta B^2 + \frac{H}{1-H}\delta A^2\right] = \frac{G_N M\hbar}{5\pi c R^2} + O\!\left(\frac{G_N^2 M^2\hbar}{c^3 R^3}\right).} \tag{100}$$

The equality of the leading term uses the specified Boulware state, large R, and fixed asymptotic ADM mass, and is a reconstruction of a known semiclassical metric rather than a distinct quantum prediction of the signed representation. The last O term denotes higher classical-field powers multiplying this one-loop contribution.

Quantum backreaction and the asymptotic charge. Chapter 4's specified massless-Dirac quantum corrections change the exterior radial and temporal metric amplitudes at order $\delta A^2, \delta B^2 = O(\hbar/r^3)$ with fixed asymptotic ADM mass M. Their first spatial derivatives decay as $O(\hbar/r^4)$, so they contribute $O(\hbar/r^2)$ to an ADM surface integral and therefore no shift to its limit at infinity. Quantum matter still changes finite-radius gravitational observables and energies. This conclusion is specific to the named fixed-mass asymptotic state; a different quantum state may also change the global mass integration constant through its physical source and boundary conditions.

18.6 The first quasilocal correction and its meaning

Chapter 6's positive exterior static gradient contribution is $E_{\rm grad}(r > R) = G_N M^2/(2R)$, while the same Schwarzschild geometry on its conventional static spatial slice has the flat-referenced Brown–York quasilocal energy

$$E_{\rm BY}(R) = \frac{c^4 R}{G_N}\left[1 - \sqrt{1 - \frac{2G_N M}{c^2 R}}\right] = Mc^2 + \frac{G_N M^2}{2R} + O\left(\frac{G_N^2 M^3}{c^2 R^2}\right). \tag{101}$$

Therefore $E_{\rm BY}(R) - Mc^2 = E_{\rm grad}(r > R) + O(R^{-2})$. This is a *leading weak-field coefficient match*, not equality of local density or of quasilocal prescriptions: the Brown–York quantity uses a specified reference and spatial slicing, while $E_{\rm grad}$ is part of an auxiliary static functional. Binding energy is negative only after the matter–field interaction is included.

18.7 Consequence for the interpretation of QGD

In any purely metric-dependent QGD sector on which the metric map is regular and boundary-compatible, the covariant gravitational Hamiltonian/Noether surface charges are pulled back from those of the same metric theory. ADM and Bondi charges therefore agree with the metric comparator when the same physical geometry and boundary data are used. The signed representation alone cannot remove GR's general localization ambiguity; this is an exact limitation of the presently accepted metric-only framework, not an obstacle to using its valid energy balance and flux formulae. The reference coframe may help organize calculations, but a unique new local energy requires a physically selected frame/observer and a justified invariant prescription (or new sector dynamics).

Relevant references: Brown–York, *Phys. Rev. D* 47 (1993) 1407, https://doi.org/10.1103/PhysRevD.47.1407; Iyer–Wald, *Phys. Rev. D* 50 (1994) 846, https://doi.org/10.1103/PhysRevD.50.846; Szabados, *Living Reviews in Relativity* 7 (2004), https://doi.org/10.12942/lrr-2004-4.

19 Quantum energy hierarchy and the limits of the natural-invariant energy dictionary (QGD–140)

Scope. Retain the complete founding master metric, $g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}]$, and the preferred Option C variational equation. Calculations here use the *previously established* metric-only, fixed-reference, $\lambda\mathcal{Q} = 0$ semiclassical branch. No independent quantum energy of the historical $\mathcal{Q}_{ab}$ is claimed or added to the metric effective action.

19.1 Two independent perturbative parameters

For an asymptotically flat source of fixed ADM mass M, set

$$u(r) := \frac{G_N M}{c^2 r}, \qquad w(r) := \frac{G_N \hbar}{c^3 r^2} = \frac{\ell_P^2}{r^2}. \tag{102}$$

Terms of order u^n can be purely classical; the source-specific Dirac vacuum polarization below is of order w relative to the leading Newtonian force. Mixed uw corrections are not automatically fixed by knowing the leading far-field w term.

19.2 Massless Dirac vacuum energy in a fixed exterior state

Adopt the specific renormalized asymptotically empty (Boulware) state of one free massless four-component Dirac field and the far-field semiclassical metric in Section 18. The existing Chapter 4 calculation yields the leading areal-radius Misner–Sharp mass, with $G_N, c, \hbar$ restored,

$$m_{\rm MS}(r) = M + \delta m_1(r) + \cdots, \qquad \delta m_1(r) = \frac{G_N M \hbar}{5\pi c^3 r^2}. \tag{103}$$

Using the spherical Einstein constraint $\mathrm{d}m/\mathrm{d}r = 4\pi r^2 \rho_E/c^2$ (where ρ_E is local static-frame energy density), its *renormalized quantum matter* contribution is

$$\boxed{\rho_E^{(Q)}(r) = \frac{c^2}{4\pi r^2}\frac{\mathrm{d}\delta m_1}{\mathrm{d}r} = -\frac{G_N M \hbar}{10\pi^2 c r^5} + \cdots.} \tag{104}$$

Integrating the leading exterior tail gives the quantum matter energy outside an areal sphere of radius R:

$$\boxed{E_Q^{\rm out}(R) = \int_R^\infty 4\pi r^2 \rho_E^{(Q)}(r)\,\mathrm{d}r = -\frac{G_N M \hbar}{5\pi c R^2} + \cdots = -c^2 \delta m_1(R) + \cdots.} \tag{105}$$

The leading quantum matter energy is negative in this *specified* state. This is not a negative energy density assigned to a fundamental σ condensate or an assertion about arbitrary quantum states. At fixed ADM mass M, the finite-radius enclosed mass is correspondingly larger: $c^2[m_{\rm MS}(R) - M] = -E_Q^{\rm out}(R)$ at this order.

19.3 Quasilocal energy, redshift and the first formal second-order term

On the static-orthogonal, flat-subtracted Brown–York convention of Chapter 6, let

$$E_{\rm BY}(R; m) = \frac{c^4 R}{G_N}\left[1 - \sqrt{1 - \frac{2G_N m(R)}{c^2 R}}\right], \qquad f_0(R) = 1 - \frac{2G_N M}{c^2 R} > 0. \tag{106}$$

Treat $m = M + \delta m_1 + \delta m_2 + \ldots$ with $\delta m_n = O(\hbar^n)$ as an *order-counting definition*. Taylor expansion about M gives the exact algebraic identities

$$\delta E_{\rm BY}^{(1)}(R) = \frac{c^2 \delta m_1(R)}{\sqrt{f_0(R)}}, \tag{107}$$

$$\delta E_{\rm BY}^{(2)}(R) = \frac{c^2 \delta m_2(R)}{\sqrt{f_0(R)}} + \frac{G_N[\delta m_1(R)]^2}{2R f_0(R)^{3/2}}. \tag{108}$$

The leading known term is therefore

$$\delta E_{\rm BY}^{(1)}(R) = \frac{G_N M \hbar}{5\pi c R^2} + \text{uncomputed subleading one-loop terms.} \tag{109}$$

The $f_0^{-1/2}$ factor in Eq. (107) is a legitimate *kinematical* response to a specified $\delta m_1(R)$; expanding that factor does not determine the next far-field physical coefficient, because omitted $O(\hbar r^{-3})$ contributions to δm_1 contribute at the same order. Likewise the explicitly positive term quadratic in δm_1 in Eq. (108) is only part of the physical order-$\hbar^2$ correction: δm_2 remains unknown. Obtaining δm_2 requires two-loop matter input and/or the self-consistent first-loop response to the perturbed geometry, including metric nonlinearities, counterterms and state conditions.

19.4 A genuine second-order field-equation recursion

For clarity use the expansion $g = g_0 + \hbar h_1 + \hbar^2 h_2 + \cdots$, $T_q[g] = \hbar\tau_1[g] + \hbar^2\tau_2[g] + \cdots$. Let $\mathcal{F}[g]$ be the Einstein–classical-matter residual, $\mathcal{F}[g_0] = 0$, and $\kappa = 8\pi G_N/c^4$. With $\mathcal{L} := D\mathcal{F}[g_0]$, expansion gives

$$\mathcal{L}[h_1] = \kappa\tau_1[g_0], \tag{110}$$

$$\mathcal{L}[h_2] = \kappa\big(\tau_2[g_0] + D\tau_1[g_0]\cdot h_1\big) - \tfrac{1}{2}D^2\mathcal{F}[g_0](h_1, h_1). \tag{111}$$

These are perturbative identities under the stated expansion, not a calculation of τ_2. They distinguish the induced $\hbar^2$ backreaction from a unique new fundamental QGD operator. For genuinely quantum gravitational loops one must additionally specify and quantize the relevant physical gravitational variables; the current accepted action includes matter loops and does not supply such a calculation.

19.5 Meaning for QGD and gravitational energy

The same $\sigma \mapsto g$ representation maps a corrected metric to finite-radius energy observables, but the *metric-only* QGD field equations have the same ADM, Misner–Sharp and Brown–York observables as their semiclassical GR counterparts in matched gauges, states and boundary conditions. The projected Option C terms are not individually conserved energy channels. The static Chapter 5 gradient energy $G_N M^2/(2R)$ is positive, the on-shell two-body binding energy negative, and the present exterior Dirac vacuum energy negative: they refer to *different functionals and sources* and therefore do not contradict one another. A distinct QGD energy prediction demands a new physically specified sector-sensitive observable or dynamics, not an arbitrary further power of $1/r$ or $\hbar$.

A QGD-specific charge-matching question. The proposed flat-space microscopic starting point has a Noether Hamiltonian H_{micro}, while an asymptotically flat isolated solution of the effective metric theory has an ADM Hamiltonian $H_{\text{ADM}} = M_{\text{ADM}}c^2$. A microscopic collective-phase model could be required to establish the *conditional matching* $\langle H_{\text{micro}}\rangle_{\text{physical,ren}} \stackrel{?}{=} H_{\text{ADM}}$, with the same reference energy, coarse-graining, source and boundary prescription. This is a proposed test of QGD's physical interpretation, not an identity proved by signed factorization or by the Newtonian flux result.

External benchmarks: E. D. Carlson et al., *Phys. Rev. D* **82**, 124070 (2010), https://doi.org/10.1103/PhysRevD.82.124070; J. D. Brown and J. W. York, *Phys. Rev. D* **47**, 1407 (1993), https://doi.org/10.1103/PhysRevD.47.1407; J. F. Donoghue, *Phys. Rev. D* **50**, 3874 (1994), https://doi.org/10.1103/PhysRevD.50.3874.

20 The historical three-channel equation revisited (QGD–141)

Purpose and provenance. The archival field-equations chapter proposed $S_\mu = Q_\mu + G_\mu + T_\mu$ with $S_\mu = \Box_g\sigma_\mu$, and also included trial higher-derivative terms. QGD–052 demonstrated that the historical Q_μ identity, T_μ coefficient, and general derivation of G_μ do not follow from the stated action. We therefore do not revive $\Box_g\sigma_\mu$ as an established independent operator. The

architecture survives in an *action-defined* successor, now including the renormalized quantum matter functional.

20.1 Exact three-channel construction using the full master metric

Use the *complete* founding map (Eq. 24), $g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}]$, with a fixed reference coframe and a specified differentiable $\mathcal{Q}_{ab}[\sigma; T]$ when $\lambda \neq 0$. Separate the adopted renormalized metric-only action, after fixing a renormalization scheme, as

$$\Gamma^{\rm ren}_{\rm metric} = \underbrace{(S_{\rm EH} + S_{\rm GHY})}_{\Gamma_{\mathsf{G}}} + \underbrace{S^{\rm cl}_{\rm F1}}_{\Gamma_{\mathsf{T}}} + \underbrace{\big(\hbar \Gamma^{(1),\rm ren}_{\rm matter} + O(\hbar^2)\big)}_{\Gamma_{\mathsf{Q}}}, \qquad \Delta\Gamma_{\rm sector} = 0. \tag{112}$$

For $X \in \{\mathsf{G}, \mathsf{T}, \mathsf{Q}\}$, define $\Pi_X^{ab} := T^a{}_\mu T^b{}_\nu (\sqrt{-g})^{-1} \delta\Gamma_X / \delta g_{\mu\nu}$. Then each channel has an exact full-map Euler derivative

$$\boxed{\mathscr{X}^{\ a}_{(s)} := \frac{1}{\sqrt{-g}} \frac{\delta \Gamma_X[g(\sigma)]}{\delta \sigma_a^{(s)}} = -2\varepsilon_s \Pi_X^{ab} \sigma_b^{(s)} - \lambda \big[(D_s \mathcal{Q})^*_{\sqrt{-g}} \Pi_X \big]^a.} \tag{113}$$

Linearity of functional differentiation proves the exact *stationarity balance*

$$\boxed{\mathscr{S}^{\ a}_{(s)} := \frac{1}{\sqrt{-g}} \frac{\delta \Gamma^{\rm ren}_{\rm metric}}{\delta \sigma_a^{(s)}} = \mathscr{G}^{\ a}_{(s)} + \mathscr{T}^{\ a}_{(s)} + \mathscr{Q}^{\ a}_{(s)} = 0.} \tag{114}$$

Unlike historical $S_\mu = \Box_g \sigma_\mu$, this is an *Euler-residual equation* and not, without additional dynamics, a free covector wave equation. The quantum channel $\mathscr{Q}_{(s)}$ means the *one-loop/higher-loop matter action contribution*; it must *not* be conflated with the separate historical metric operator $\mathcal{Q}_{ab}$. This is an explicitly *updated naming convention*: the earlier QGD–098 decomposition reserved $\mathscr{Q}$ for a derivative-map contribution and set it to zero in the algebraic metric sector. That earlier symbol is better called $\mathscr{D}_{\mathcal{Q}}$ here. Its effect is already carried by the $-\lambda(D_s\mathcal{Q})^*$ term in *each* of the three channels above. The old and new conventions must not be added together or counted twice.

In the fixed-coframe, $\lambda = 0$ specialization and $c = 1$, write $\mathcal{R}^{ab} := T^a{}_\mu T^b{}_\nu (G^{\mu\nu} + \Lambda g^{\mu\nu})$ and use similarly $T^{ab}_{\rm cl/q}$ for the frame-projected stress. Then

$$\boxed{\begin{aligned} \mathscr{G}^{\ a}_{(s)} &= -\frac{s_s}{8\pi G_N} \mathcal{R}^{ab} \sigma_b^{(s)}, \\ \mathscr{T}^{\ a}_{(s)} &= s_s T^{ab}_{\rm cl} \sigma_b^{(s)}, \\ \mathscr{Q}^{\ a}_{(s)} &= s_s T^{ab}_{\rm q} \sigma_b^{(s)}, \\ \mathscr{S}^{\ a}_{(s)} &= \mathscr{G}^{\ a}_{(s)} + \mathscr{T}^{\ a}_{(s)} + \mathscr{Q}^{\ a}_{(s)} = 0. \end{aligned}} \tag{115}$$

Compact natural-invariant projection of the Einstein residual. Equivalently, adding the three channels in Eq. (115) gives the concise field equation

$$\boxed{s_s \sigma_b^{(s)} \left[-\frac{\mathcal{R}^{ab}}{8\pi G_N} + T^{ab}_{\rm cl} + T^{ab}_{\rm q} \right] = 0, \qquad (s = 1, \ldots, N).} \tag{116}$$

This is the exact action-derived *projected Einstein residual* in the stated fixed-coframe, $\lambda\mathcal{Q} = 0$, metric-only specialization, in units $c = 1$. The curvature symbol $\mathcal{R}^{ab}$ is the frame pullback of the Einstein tensor *plus* the cosmological term, not the Ricci tensor alone. Under a regular surjective metric map, the full collection of sector equations is equivalent to the semiclassical Einstein equation; on singular or rank-deficient sector branches, vanishing projections need not imply

vanishing of the full Einstein residual. For a nonzero specified derivative slot $\lambda\mathcal{Q}_{ab}$, Eq. (116) is *not* the complete Euler equation: the volume-weighted functional-adjoint contribution in Eq. (29) must also be retained. Restoring units replaces $8\pi G_N$ with $8\pi G_N/c^4$ in the Einstein source equation. For the full derivative map, the $-\lambda(D_s\mathcal{Q})^*$ term acts on *each* Π_X; it is not an additional independently propagating quantum source. This reproduces Option C, because $2s_s\Pi^{ab}\sigma_b^{(s)} = \lambda[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi]^a$ is exactly Eq. (114) rearranged.

Full-master-metric extension: the projected Einstein residual with derivative response. The compact projection in Eq. (116) is valid only for $\lambda\mathcal{Q} = 0$. For a specified differentiable $\mathcal{Q}_{ab}[\sigma; T]$, the *complete* stationarity condition is instead obtained by retaining its functional-adjoint term. Define $\Pi^{ab} := \Pi^{ab}_{\mathsf{G}} + \Pi^{ab}_{\mathsf{T}} + \Pi^{ab}_{\mathsf{Q}}$, where the metric Euler tensors Π^{ab}_X are defined in Eq. (113); in $c = 1$ units this is $\Pi^{ab} = \frac{1}{2}[-\mathcal{R}^{ab}/(8\pi G_N) + T^{ab}_{\rm cl} + T^{ab}_{\rm q}]$. The Option-C form of the complete natural-invariant equation is therefore

$$\boxed{\begin{aligned} &s_s\sigma_b^{(s)}\left[-\frac{\mathcal{R}^{ab}}{8\pi G_N} + T^{ab}_{\rm cl} + T^{ab}_{\rm q}\right] \\ &\qquad = \lambda\left[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\right]^a . \end{aligned}} \tag{117}$$

Here $s_s = -\varepsilon_s$, $D_s\mathcal{Q}$ is the Fréchet variation with respect to $\sigma^{(s)}$ at fixed coframe, and $\mathcal{R}^{ab} := T^a{}_\mu T^b{}_\nu(G^{\mu\nu} + \Lambda g^{\mu\nu})$ is *not* the Ricci tensor. Both sides arise by varying the same *metric-only* renormalized action through the full composite metric; the right side does not constitute an independently postulated source. The matter-loop contribution $T^{ab}_{\rm q}$ and the historical metric operator $\mathcal{Q}_{ab}$ remain distinct. A nonzero λ is not microscopically predicted by this rewriting, and physical equivalence to the metric field equations requires an admissible full-rank metric map. Setting $\lambda = 0$ returns Eq. (116) without an extra assumption.

Proposition 20.1 (Regular full-rank correspondence and its limit)**.** *For a metric-only action on a regular branch where the full $D_\sigma g$ is surjective on admissible metric variations, $\mathscr{S}^{\,a}_{(s)} = 0$ for all sectors is equivalent to $\mathcal{E}^{\mu\nu}_g = 0$. The sum of the three channels is then zero; the channels individually need not vanish. There is no independent σ-wave energy or new force coefficient obtained just by splitting the zero residual.*

Proof. Use the complete functional chain rule $\delta\Gamma = \int \sqrt{-g}\,\mathcal{E}^{\mu\nu}_g(D_\sigma g)_{\mu\nu}\delta\sigma$. Surjectivity gives the converse; linearity gives the channel identity. □

20.2 The balance is not an independent local energy-conservation law

The geometric Bianchi identity and renormalized matter Ward identity imply, on the matter equations and with appropriate state-dependent backreaction, $\nabla_\mu(T^{\mu\nu}_{\rm cl} + T^{\mu\nu}_{\rm q}) = 0$; they do not imply separate conservation of every channel. Microscopic reference-space Noether conservation (Chapter 6), the projected Euler equation, and ADM/Bondi/quasilocal energies are *three different constructions*. This distinction prevents reading $\mathscr{G}, \mathscr{T}, \mathscr{Q}$ as three gauge-invariant local energy densities.

Renormalization caveat. A finite local counterterm may be moved between Γ_{G} and Γ_{Q}, with an equal and opposite change, leaving $\Gamma_{\rm metric}$ and $\mathscr{S}$ unchanged. The individual named channels therefore require a declared subtraction convention; only the total field equation is scheme independent after physical parameters are matched.

20.3 A nontrivial test of both quantum energy and pressure

For the published free massless-Dirac Boulware far field of Section 13, use $G_N = c = \hbar = 1$ and $\langle T^\mu{}_\nu\rangle = M\,\mathrm{diag}(4, 2, -3, -3)/(40\pi^2 r^5)$. The effective energy density and radial pressure

are respectively $\epsilon_q = -M/(10\pi^2 r^5)$ and $p_{r,q} = M/(20\pi^2 r^5)$. The spherical Einsteinian source equations give, at leading quantum far-field order,

$$\boxed{\begin{aligned} &\delta m'(r) = 4\pi r^2 \epsilon_q = -\frac{2M}{5\pi r^3}, && \delta m(r) = \frac{M}{5\pi r^2}, \\ &\frac{\mathrm{d}}{\mathrm{d}r}\ln D = \frac{8\pi r(\epsilon_q + p_{r,q})}{1-2m/r} = -\frac{2M}{5\pi r^4} + \text{higher orders}, && \ln D = \frac{2M}{15\pi r^3} + \text{higher orders}, \end{aligned}} \tag{118}$$

where $D = 1 + B^2 - A^2$. The first equation measures the *enclosed energy* response; the second responds to *energy density plus radial pressure*, and explains why $A^2 \neq B^2$ for this quantum state. Both arise from the same $T_q^{\mu\nu}$ and the same Euler balance, rather than independent historical Q_μ and G_μ source guesses.

Conclusion. The historical formula retains conceptual value as an architecture of gravity, matter and quantum response. Its justified successor is Eq. (114), not $\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu$ unless a new, properly normalized and stable independent covector kinetic action is separately established. This is a completed action-based reinterpretation, not a microscopic proof of the historical wave law.